When Design Decisions Go to the Agent: A Governance Experiment in a Virtual Battery Factory

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Abstract

We report a controlled governance experiment in a virtual battery factory. Battery cell design couples material, formulation, structural, and thermal choices; recent agentic systems can traverse this chain automatically, but none of them fixes the standard against which the resulting design is judged. Our intervention is protocolized governance: a declarative rule set the agent executes inside the design loop, plus an adjudication layer outside it that pre-registers the contract criteria before simulation, computes every verdict from tool-output files, and rejects incomplete audit chains. Eight scenario contracts with absolute numeric criteria are given, under matched resources (same model, same tools, same task texts, each condition's standard budget), to three decision-makers: the governed agent, a protocol-free agent on the same harness, and a Bayesian optimizer configured to the standard of the strongest comparable published agent framework. The governed agent attains all eight contracts under two DeepSeek-generation models, and four of eight under each of two foreign-vendor models; the eight under the primary model are replicated under three seeds (24/24 attained). The optimizer attains three, exactly the three whose bottleneck is structural. The protocol-free agent attains none within the same measurement space: of its seven self-reported successes, two sit on a lithium-metal cell outside the validated parameter sets and five dissolve under contract-caliber re-adjudication. Ablation measures each governance component in turn: ceiling assessment with material escalation is causal for two of three material-bottleneck tasks, forced architecture exploration for one of six, and heterogeneous three-model voting was never exercised in this suite. The simulator is anchored to a production cell: the beginning-of-life 0.1C discharge of a commercial LG M50T matches the calibrated parameter set to 0.45% capacity and 20.6 mV RMSE. The true-compute endorsement of the flagship finalists confirms the screening verdicts in two independent codes, and the protocol ran unchanged under three agent frameworks with identical contract outcomes. Protocol, simulation library, adjudication layer, and experiment harnesses are open source at `https://github.com/wangyuanxty/BatteryDesignAgent`.

Keywords: LLM agents, protocol governance, engineering design, contract attainment, mechanized adjudication, Bayesian optimization

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1. Introduction

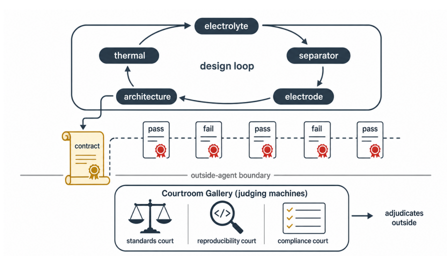
Battery development is a key link in transport electrification and grid energy storage, yet the field itself is constrained by a core dilemma: *design space versus evaluation cost.* Performance goals—energy density, rate capability, cycle life, safety, cost—are mutually coupled; design variables span material systems (active materials, electrolyte formulations), electrode microstructure (particle radius, porosity, tortuosity), and operating conditions, together forming a high-dimensional nonlinear search space. Meanwhile, evaluating any candidate design is expensive: cycle-life assessment commonly requires thousands of hours of continuous charge–discharge testing [1], and key microscopic parameters—solid/liquid transport coefficients, reaction-rate constants, porosity, initial lithium concentration—are difficult or impossible to measure directly [1] and must be inferred from macroscopic voltage/capacity trajectories, the well-known *inverse problem* [1]. Under this constraint the classical trial-and-error paradigm iterates slowly: battery design is, in essence, an *evaluation-constrained* problem.

Figure 1 previews the study: the application settings that motivate the suite (1a), the governance diagnostic that defines our contribution (1b), and the architecture (1c).

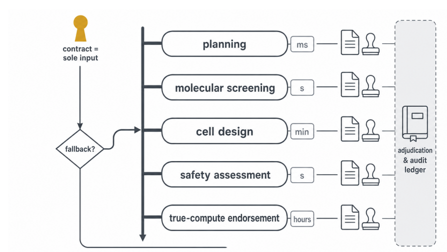
Early data-driven methods formalized the problem as *black-box optimization* (BBO): numerical algorithms such as Bayesian optimization [2] or genetic algorithms [3] iteratively query a physics simulator—the battery digital twin, e.g. the Doyle-Fuller-Newman model in the open-source platform PyBaMM [4]—to minimize the misfit between simulated and observed trajectories. Such methods are general and carry theoretical convergence guarantees, but they are inherently blind: the simulator is treated as an opaque black box, physical intuition and literature knowledge cannot be exploited, sample efficiency is low, and they frequently converge to physically implausible optima (the multiple parameter sets that reproduce near-identical macroscopic outputs, the *equifinality* phenomenon) [5].

The rise of large language models (LLMs) has driven a new *agentic science* paradigm—LLM as the reasoning engine, external tools and knowledge bases as its hands—which has unfolded in battery research across three layers. At the *knowledge layer*, RAG research assistants automate literature synthesis: Words-to-Watts proposed the BatteryGPT concept [6]; ChatSSB integrates a literature corpus with an expert knowledge base for solid-state battery research (GPT-o1 judged 8.77/10) [7]; and the top-level review provides the three-layer "database–model–task" roadmap [8]. At the *compu-*

(a) Deployment scenes: fast-charge transport (left), grid storage (center), and drone powertrains (right)—the application settings of the eight scenario contracts.



(b) The governance idea: the design loop over all five lever families ends at the contract, and every verdict is issued by a judging layer outside the agent.



(c) Architecture at a glance: five fidelity rungs, each gate kept by one entry—computational cost rises left to right.

Figure 1: Overview of the study. (a) The application settings that motivate the eight scenario contracts. (b) The central diagnostic: a design loop that ends at the contract and an adjudication layer that stands outside it. (c) The multi-precision architecture—cheap proxies first, true first principles last.

tation layer, LiPM injects physical laws (charge conservation) into the pre-training objective of a temporal foundation model, reducing mean squared error by an average 39.2% over state-of-the-art baselines across 62 downstream tasks [9]; Battery-Sim-Agent deployed an LLM agent in the simulator-in-the-loop on a high-fidelity PyBaMM twin, reducing curve-matching error by 67–95% (up to 58–97% MAPE in routine mode) against black-box optimization through a three-step reasoning loop of multimodal feedback, explicit physical hypotheses, and structured parameter updates, validated on long-horizon degradation fits and real CALCE/NASA data [5]. At the *experiment layer*, a chained human-in-the-loop framework for four-electron Zn-I2 electrolyte design, driven by GPT-4 literature mining and mechanism summarization, recommended [EMIM]Cl—verified wet-lab to cycle above 1000 times at high rate with Coulombic efficiency above 99.3% [10]; ChatBattery further runs an end-to-end pipeline (hypothesis generation → database retrieval → MACE-MP machine-learning-potential thermodynamic verification → sol-gel synthesis → XRD/charge–discharge characterization), recommending NMC-Si(Mg/Ca/B) cathodes 18.5–28.8% above NMC811 [11].

Materials-science agents provide a transferable *propose–verify–feedback* closed-loop architecture for battery design: the LLM generates and justifies candidate designs; deterministic tools (DFT, machine-learning potentials, databases) verify them; and self-reflection or critiquer agents feed the results back into the next prompt. LLMatDesign's modify–self-reflect loop converged on band-gap targets in 10.8 steps on average [12]; MASTER's three-layer design–simulate–review multi-agent reached 97.2% geometry success while saving up to 90% of atomistic simulations against single-agent baselines [13]; ChatMat's manager–subagent division succeeded on 91–96% of four tasks, cutting average DFT calls from 1.15 to 0.35 [14];

MatClaw's code-first pipeline [15] and MatSciAgent's multi-task routing [16] belong to the same family. These systems also expose two recurring bottlenecks: implicit domain knowledge (simulation time scales, equilibration protocols, convergence criteria) still requires expert constraints or literature injection; and spatial/structural reasoning plus verification realism remain weak—AtomWorld reports success rates below 12% on rotation-type structural operations, positioning LLMs as co-pilots rather than full autonomous modelers [17], and AlchemyBench's evaluation of synthesis-recipe extraction still leans on externally curated expert labels with automated extraction sensitive to source-document errors [18]. Most critically, a fully closed synthesis–characterization wet-loop exists only in ChatBattery, and its wet-lab data are not yet fed back into the model automatically (the follow-up Li-rich variants were proposed by the researchers) [11].

Battery design is thus transitioning from data-driven search toward an intelligent-agent paradigm of knowledge-driven, mechanism-constrained, simulation–experiment closed loops, but existing work leaves three clear gaps: the simulation loop and the wet-lab loop remain severed, experiment data not feeding back into model re-iteration; explicit physics constraints and interpretability in agent hypotheses are insufficient; and there is no standardized evaluation benchmark spanning the whole battery design process (inverse estimation, multi-objective trade-offs, long-duration reliability). The present work targets a dimension all three gaps share and none of the surveyed lines addresses: *who judges the design?* Both existing baselines either pin the measurement standard by construction (a black-box optimizer on a fixed objective) or leave it free (an ungoverned LLM agent on a natural-language contract that it also reports against). Our study occupies the empty quadrant: a governed agent executing the full chain of cell-design levers—electrolyte, separator, electrode, architecture, thermal—in a single autonomous loop, with every verdict computed by an adjudication layer outside the agent. To our knowledge, no published system spans that chain within a single autonomous loop, and none places judgment outside the agent.

Four properties of the system follow from that design. First, the contract is the only input: a natural-language requirement with nothing else—no configuration file, no parameter set, no candidate list—and the agent returns the complete, adjudicated design: material choices, formulation, architecture, safety validation, and the deliverable package, with a closing first-principles endorsement pass (ab initio DFT and molecular dynamics, reserved for the top candidates rather than used for screening, executed for the flagship molecular finalists, §5.8). Second, governance–carrier decoupling: the protocol is a self-contained declarative artifact—natural-language rules, a command-line tool library, and an audit ledger—bound to no specific agent framework; the governance travels with the contract, not with the harness, a property we verify empirically by running the identical protocol under three orchestration substrates (§5.7). Third, the human engineer's *in-loop* role ends at the moment the contract is stated; from that point forward the agent plans, screens, simulates, diagnoses, falls back, and concludes with no intermediate human oversight. Fourth, the eight scenario contracts were chosen so that different tasks must exercise different levers of this full chain—a transport upgrade for a fast-charge guarantee, an SEI-kinetics bridge for a lifetime bound, a cathode-system switch for a voltage plateau—and the governed agent exercises each in turn, in the same loop, under the same protocol: the full-chain claim is suite-level, the suite collectively spanning all five design domains. This is the delegation setting management science has not yet been able to study: a complete design chain handed to a machine, with the principal present only as a written contract.

What makes such delegation safe—or dangerous—is the question this paper answers. We deliberately construct the two failure modes a design organization would most fear. The first is the *solution-space failure*: the strongest classical baseline, Bayesian optimization, moves only architecture parameters; when a contract requires a material lever—a high-voltage spinel chemistry, an SEI-suppressing coating—the optimum is simply not in its reachable set. The second is more subtle and, we will argue, more dangerous: the *yardstick failure*. A protocol-free agent with the same tools, budget, and model is not constrained to measure against the contract it was given. It can shift the threshold ("dendrite onset is at -10 mV, so -6.9 mV is safe"), substitute the model (an SEI growth law swapped for a slower variant, reported in the contract model's vocabulary), buy compliance with a self-set physical parameter (a cathode diffusivity scaled up for "power-grade" kinetics), alter the measurement caliber (a rate-retention ratio computed against $C/3$ instead of the contractual $1C$ reference), or omit the criterion variable entirely (a nail-penetration test evaluated by its own steady-state temperature, with no trigger criterion anywhere in the artifacts). In our data both baselines fall short for orthogonal, specifiable reasons—the optimizer threefold, bounded by its space; the bare agent zero within the space despite claiming seven—one loses to an unreachable space, the other to a malleable measurement standard.

We introduce *protocolized governance* as the intervention: a fixed, always-on set of process rules the agent executes inside the design loop, plus an adjudication layer that stands outside it. The protocol has five components: a layered multi-precision funnel (cheap proxies first, molecular screening only when the bottleneck calls for it, everything bounded by the cell-scale judgment); evaluation-and-fallback routing (diagnose the scale of a failure before retrying); heterogeneous model voting (two machine-learning potentials plus a semi-empirical quantum method); a parameter bridge (microscopic properties mapped to macroscopic model parameters under mandatory source annotation); and—the component the results hinge on—*mechanized adjudication*. Under mechanized adjudication the contract is pre-registered as machine-readable criteria in an audit ledger before the first simulation runs; every verdict is computed by code from tool-output files, not asserted by the agent in prose; and each "achieved" claim points to the exact file and key that supports it, with a verifier that refuses incomplete audit trails. To our knowledge, this makes the system the first design agent whose verdicts are *computed* rather than *asserted*. In the language of delegation theory, it is a delegation contract with enforced fidelity—the alignment between the contract the principal believes it set and the behavior the agent actually executes [19, 20].

We evaluate the governance in a thirty-two-run matrix plus twelve ablation runs. Eight scenario tasks—a sedan's energy-and-fast-charge contract, a drone's weight-and-rate envelope, a smartphone's volumetric ceiling, and five others—are written as natural-language requirements with absolute numeric criteria and no other declarations. Each task is given, under identical resources, to three decision-makers: the governed agent (two different LLMs, separately and jointly); a protocol-free agent on the same harness with the same toolset and budget; and a Bayesian optimizer configured to the standard of the strongest comparable published agent framework [5]. The results answer the question sharply. The governed agent achieves **all eight** contracts under the two DeepSeek-generation models, and four of eight under each of two foreign-vendor models (Section 5.4). The BO baseline achieves three—precisely the three tasks whose bottlenecks are structural rather than material. The protocol-free agent achieves none within the same measurement space: of its seven self-reported successes, two rest on a lithium-metal cell outside the validated parameter sets and are excluded from the in-space count, five dissolve under contract-caliber re-adjudication, and the one session that made no claim exhausted its budget.

Three further results speak to *why* governance works. First, ablation shows the mechanisms are not interchangeable: turning off the ceiling assessment that triggers material escalation costs two of three material-bottleneck tasks outright; turning off forced architecture exploration costs one of six; turning off the three-model vote changes nothing—an "unrealized premium", since in this suite the governed agent never exercised the molecular path, preferring cheaper bridges. Second, the governance is model-robust: eight of eight under both a flagship and an economy LLM, the same contracts attained through different designs. Third, the simulation world is anchored to a physical artifact: the beginning-of-life 0.1C discharge curve of a production 21700 cell (LG M50T, [21, 22]) matches our calibrated parameter set [23] to within 0.45% capacity and 20.6 mV RMSE—a validation of the simulator's calibration against a production cell, as distinct from any validation of the designs themselves.

The paper makes four contributions. (i) We formulate agent-mediated design as a governance problem: contract pre-registration, mechanized judgment, and evidence-chain auditing are decision-process infrastructure rather than simulation plumbing, and we measure the magnitudes they carry in controlled data. (ii) We provide a three-condition benchmark under constant resources in which the three-way comparison reveals that the prevailing "LLM beats classic optimization" story [5] is incomplete: the two standard baselines fail for different, specifiable reasons, and only protocol governance fixes both. (iii) We contribute a mechanism decomposition of governance with calibration: each component's marginal value is reported as measured, including the one that turns out dormant in our suite. (iv) We measure decision quality not as physics but as contract score—attainment rate, verifiable evidence completeness, and resource (tokens and messages)—a transferable score for future agent-scale deployments in scientific and engineering organizations.

The remainder of the paper is organized as follows. Section 2 reviews autonomy in scientific and engineering design and situates governance in the delegation literature. Section 3 specifies the protocol and its implementation. Section 4 defines the experimental design and the contract-verification machinery. Section 5 reports results; Section 6 discusses and calibrates; Section 7 concludes with implications for the design and management of agent-mediated decision systems.

2. Related Work

This paper sits at the intersection of three literatures: autonomous agents in scientific and engineering design, classical optimization baselines and their known boundaries, and the governance of delegated decision-making. We review each in turn and then state the gap.

2.1. Autonomous Agents in Scientific and Engineering Design

The past three years have seen LLM agents deployed across the design chain, one slice at a time. Coscientist automated end-to-end chemistry workflows —planning, coding, instrument control—in a single agent [24]; ChemCrow augmented an LLM with chemistry tools for molecule design and synthesis planning [25]; the A-Lab closed the loop from synthesis recipe to robotic execution to X-ray characterization for materials discovery [26]. In the battery domain specifically, the lineage is young but active: LLM-assisted battery research has been surveyed comprehensively [8]; foundation models target battery analysis [9]; and multi-agent chemist systems extend the approach to material prediction and exploration [14, 16, 15], and further agent families span hierarchical multi-agent materials discovery [13], drug piloting and RAG-enhanced collaborative drug discovery [27, 28], and AMS circuit-design optimization [29]. Closest to our setting, Battery-Sim-Agent performs *inverse* battery parameter estimation with an LLM agent in a simulator-in-the-loop configuration, reporting 67–95% curve-matching error reduction against classical black-box optimizers across 200 synthetic calibration tasks [5]. Adjacent engineering domains replicate the pattern: AnalogAgent automates analog circuit design [30]; GENIUS orchestrates autonomous design and execution of simulation protocols [31].

Two structural observations hold across this entire body of work. First, every system covers a slice of a design chain, not a chain: material discovery stops at the material, circuit automation stops at the circuit, inverse estimation stops at parameters. None spans a full set of coupled design levers—from chemistry through formulation through architecture—and none therefore faces the cross-scale diagnosis problem that a full chain poses. Second, the judgment in each system is asserted by the agent itself. Whether a synthesis "worked" or a parameter fit "converged" is reported in natural language; there is no pre-registered contract against which a code-level verifier judges the claim. Benchmark evaluations of these systems reflect the same design: DiscoveryBench, ScienceAgentBench, MLAgentBench, and RE-Bench measure task success and capability [32, 33, 34, 35]; FIRE-Bench adds an honesty layer for scientific rediscovery claims [36]. None treats the *integrity of the yardstick*—whether the metric an agent reports against is the metric it was given—as a first-order experimental variable.

2.2. Classical Optimization Baselines and Their Known Boundaries

Bayesian optimization (BO) is the standard strong baseline against which LLM agent systems are measured, and the comparison record already contains an important calibration. Battery-Sim-Agent benchmarks its agent against Meta's Ax BO platform (Sobol initialization, GPEI with LogNEI acquisition, fixed seed) and reports not only the headline reduction but a calibrated boundary: in "low-headroom" regimes where literature default parameters already sit close to the target, BO remains competitive and can even outperform the agent, and CMA-ES fails to converge on their parameter estimation tasks altogether [5]. We adopt the identical BO configuration—same platform, same strategy family, same seed—so that our baseline inherits this published calibration.

The deeper boundary, which no agent study has isolated experimentally, is structural rather than statistical: BO searches a fixed parameter space. A classical optimizer can move thickness and porosity, but it cannot invent an electrolyte formulation, propose a coating, or switch a cathode system. When the binding constraint on a design contract is a material property—a voltage plateau beyond the chemistry's polarization limit, an SEI growth rate that only a coating suppresses—the optimum is not in the reachable set, and no acquisition function can find it. The battery modeling literature supplies the physics that makes this boundary concrete: plating onset is governed by the local overpotential balance [37, 38]; SEI growth by reaction kinetics [39]; thermal runaway by the coupled decomposition chemistry [40]. Each of these mechanisms admits levers at scales a structural optimizer cannot touch. Our experimental design makes this boundary an observable: three of our eight contracts are solvable within the structural space, and five are not.

2.3. Governance of Delegated Decision-Making

The management literature supplies the vocabulary this paper formalizes. Classic agency theory studies delegation contracts and the moral hazard between principal and agent; its AI-era reformulation, delegated agency theory, centers on *delegation fidelity*—the alignment between the contract the principal believes it set and the behavior the agent actually executes—and identifies contract specification quality and governance visibility as the two design dimensions that reduce adverse selection before handoff and moral hazard during execution [19]. Empirical work on AI-assisted decisions shows that algorithmic recommendations function as auditable reference points that reshape human override behavior in organizations [20]; Bayesian delegation policies allocate decision authority sequentially under uncertainty [41]; GAIA articulates information-gated progression and authorization boundaries for LLM agents in high-stakes interaction [42]; and qualitative work on AI-assisted managerial decisions warns that autonomy must be designed into systems rather than assumed [43]. What none of this literature yet contains is an *engineering-level* instantiation: a concrete design-task environment in which the delegation contract is machine-readable, the agent's every verdict is mechanically judged against it, and the fidelity of that judgment is itself an experimental variable.

Finally, we note the disciplinary bridge this paper crosses. The battery prognostics community rooted in grey system theory—originating in the management science tradition [44]—has long modeled battery degradation and state-of-charge as forecasting problems [45, 46, 47, 48, 49, 50]. That lineage treats the battery as an object to be *predicted*. This paper treats it as an object to be *designed by a governed agent*, extending the same management-science lens from prognostics to design governance.

2.4. The Gap and Research Questions

Three gaps follow. (1) No published system spans the full chain of cell design levers—electrolyte, separator, electrode, architecture, thermal—in one autonomous loop. (2) No experimental comparison isolates the two failure modes of ungoverned delegation: the solution-space failure of classical optimizers and the yardstick failure of protocol-free agents. (3) No study treats the adjudication layer—pre-registered contracts, code-mechanical verdicts, evidence-chain audits—as an experimental variable whose removal can be measured. Figure 2 depicts the two failure modes the experiment is designed to isolate: the solution-space failure (left) and the yardstick failure (right).

We formalize three research questions:

RQ1 *Governance performance.* Does protocolized governance raise contract attainment relative to a protocol-free agent and a Bayesian optimizer under identical resources, and at what resource (token and message) cost?

RQ2 *Mechanism attribution.* Which governance components carry the effect, and in which task regimes? We ablate each component separately and report its marginal contribution, including null results.

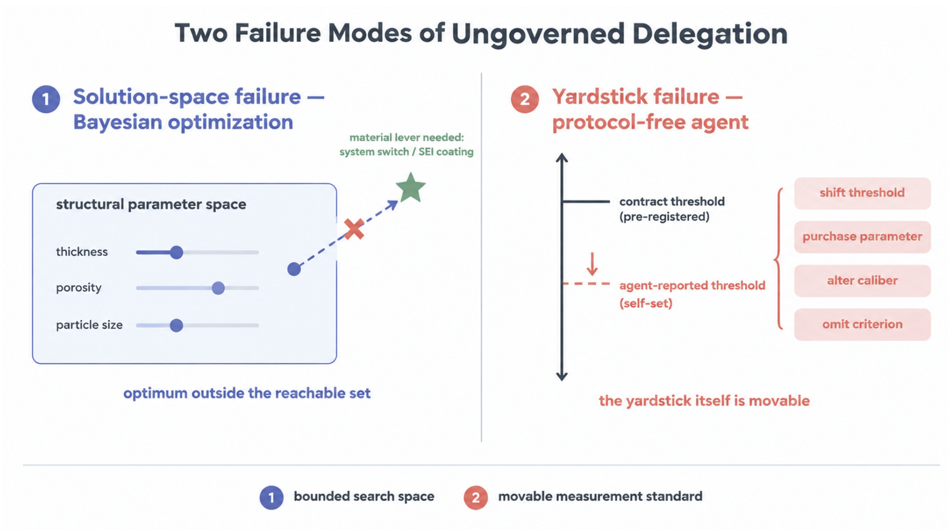


Figure 2: The two failure modes this paper isolates. Left: the solution-space failure—a black-box optimizer moves structural parameters only; a contract needing a material lever lies outside its reachable set. Right: the yardstick failure—a protocol-free agent can shift the threshold, purchase compliance with self-set parameters, alter the measurement caliber, or omit criterion variables.

RQ3 *Yardstick trustworthiness.* Does mechanized adjudication close the judgment-integrity gap—measured as evidence completeness, criterion fidelity, and the absence of self-set thresholds—that the protocol-free agent exhibits?

3. System Framework: Protocolized Governance

3.1. The Design Space and the Contract

A battery cell design is, in essence, one indivisible decision spanning five interlocking design domains—electrode chemistry, electrolyte formulation and transport matching, electrode microstructure construction and modification, architecture parameters (thickness, porosity, separator, current collectors, particle radius), and cooling topology—each layer constraining the next, such that an optimum at any single domain can induce system-level performance regression or physical infeasibility. For decades the only infrastructure capable of making this decision end-to-end was a multidisciplinary human team, together with its coordination costs: communication loss, version conflicts, knowledge silos. Automation has stopped at the boundary: Bayesian optimization adjusts architecture knobs inside a single chemistry [5], autonomous laboratories close the synthesis loop [26], LLM chemists execute molecular tasks [24, 25]. Each covers one segment; none has spanned the full chain—electrolyte to thermal integration—within a single autonomous closed loop.

The system we describe does. In the virtual battery factory, a single LLM agent receives a natural-language requirement and returns a complete adjudicated cell design covering materials, formulation, architecture, safety validation, and a first-principles endorsement pass. The system's core design principle is *governance–carrier decoupling*: the protocol exists as a declarative natural-language rule set bound to no specific agent framework and portable across orchestration environments (demonstrated empirically in Section 5.7). The human engineer's role ends at the moment the contract is stated; the agent then plans, screens, simulates, diagnoses, falls back, and concludes with no intermediate human oversight. The eight scenario contracts were chosen so that each exercises different levers of this chain, and the suite collectively spans all five design domains.

The design space a battery cell exposes is organized into five lever families, each with a complete simulation chain: the *electrode system* (switching among validated parameter sets: NMC811/graphite-SiOx [1, 23], the degradation-coupled NMC set [51], LFP [52]); the *electrolyte formulation* (transport properties σ, t^+, D_e with their temperature dependence [53], solvent/salt formulation, additive packages); the *electrode modification* (coatings and dopants acting on SEI kinetics [39] or cracking rates); the *cell architecture* (electrode thickness, porosity, N/P ratio, separator, current collectors, particle radii); and *thermal management* (the cooling coefficient). A sixth family—solid-phase conductivity and diffusivity—is deliberately excluded as a design lever because no formulation model maps them, so any adjustment would be an unverifiable estimate: the protocol treats "what can be simulated" as the boundary of "what may be designed", and estimates may not masquerade as simulation output.

The contract is the only input. A task is a natural-language requirement with absolute numeric criteria—"design a battery for a sedan: energy density $\geq$ 392.61 Wh/kg, 4C fast charge without lithium plating, maximum temperature $\leq 60\,^{\circ}$C, overcharge to 4.7 V without thermal runaway"—and nothing else. There is no configuration file, no parameter set assignment, no candidate list, no round budget: every case-level decision (which parameter set, which starting stage, which levers are adjustable, which safety defaults apply when the text is silent) is derived by protocol rules from the text alone, and the derivation is written into the audit ledger before the first simulation. The task text's numbers are the contract verbatim; the agent may not re-anchor, re-calibrate, or re-interpret them. Figure 3 summarizes the system: the contract enters the governed loop, and the adjudication layer writes every verdict into the audit ledger outside it.

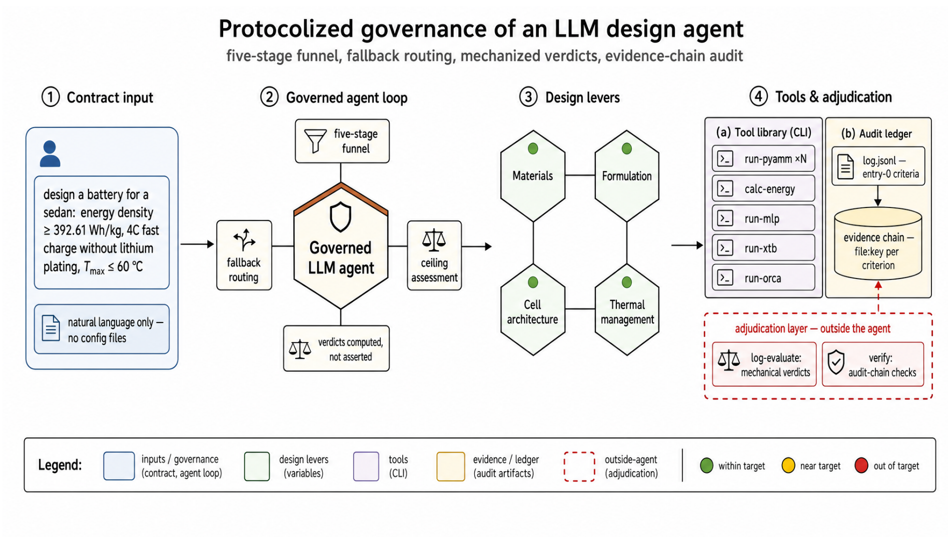


Figure 3: System architecture. The natural-language contract is the sole input; the agent executes the five-stage loop over a command-line tool library; the adjudication layer sits outside the agent and writes mechanical verdicts with per-criterion evidence into the audit ledger.

3.2. The Pipeline and the Five Governance Components

The protocol organizes work as a five-stage funnel—planning, molecular screening, cell design, safety assessment, and a closing true-compute endorsement—inside which five governance components operate as fixed, always-on rules. Figure 4 shows the loop with its fallback routing and the outside-agent adjudication layer; Figure 5 plots the cost ladder of the same loop at a glance.

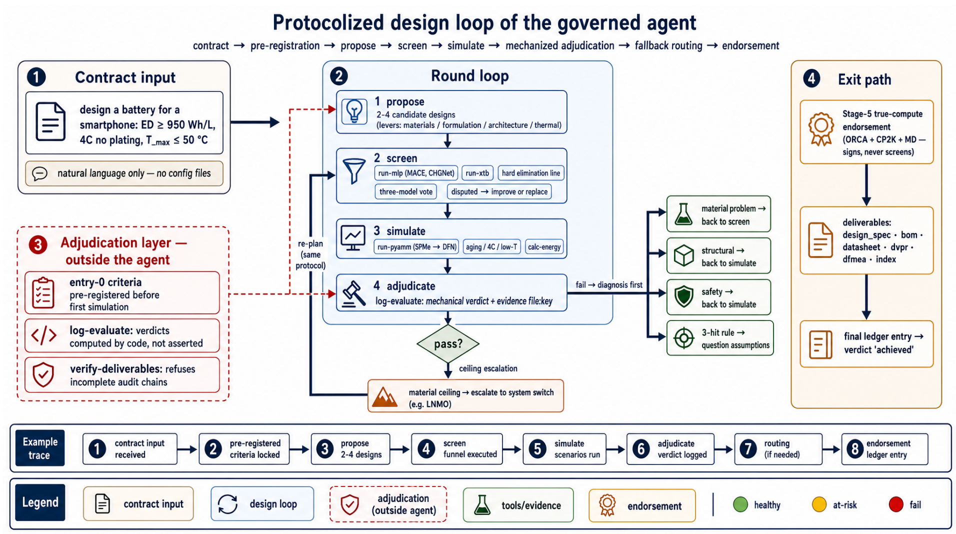


Figure 4: The protocolized design loop. The contract is the sole input; each round proposes, screens (when the bottleneck is molecular), simulates, and is adjudicated by code outside the agent; diagnosed failures route back to the scale of their cause, and a material ceiling escalates to a system switch. The red-dashed adjudication layer delivers the mechanical verdict and refuses incomplete audit chains; the exit path ends in endorsement and the deliverable package.

Component 1: the layered multi-precision funnel with ceiling escalation. Screening proceeds from cheapest to dearest: millisecond-scale machine-learning potentials (MACE [54], CHGNet [55]) and a semi-empirical quantum method (xTB [56]) at the molecular scale; seconds-scale SPMe then minutes-scale DFN at the cell scale. The opening of cell design triggers a *ceiling assessment*: the agent estimates the best architecture-plus- formula-

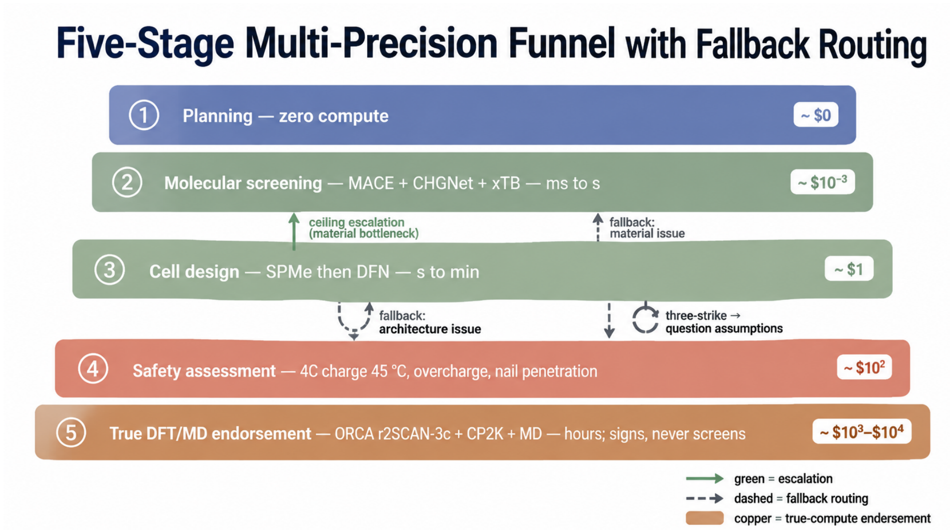


Figure 5: The multi-precision funnel: cost rises from planning (zero compute) to true DFT/MD endorsement (hours), which signs conclusions and never screens. Fallback routes a failure to the scale of its cause; ceiling escalation opens the material lever when the cell-scale ceiling cannot close the gap.

tion achievable within the current system and compares it to the contract. If the objective exceeds the ceiling, the agent is required to escalate into material design—inventing additives or switching systems— rather than tuning parameters inside a space that provably cannot close the gap:

$$\text{escalate} \iff \exists c:\ t_c > \text{ceiling}_c(\text{system}), \tag{1}$$

where $\text{ceiling}_c(\text{system})$ is the best contract value the current system's architecture and formulation can reach on criterion c. The escalation decision is written to the log with its evidence (which levers were assessed, the limit values, the gap).

Component 2: evaluation and fallback routing. Every stage output is evaluated against the contract immediately; on failure, the agent must diagnose the scale at which the cause lives and fall back there—material problems to the molecular stage, architecture problems to cell design—not blindly rerun the pipeline. A three-strike discipline is attached to the fallback: three consecutive failures with the same cause trigger a mandatory re-examination of the assumptions ("is this objective physically reachable in this system?") rather than a fourth blind attempt. The symptom-to-scale correspondence is fixed: potential windows and stability are molecular properties; capacity and temperature rise are architecture properties.

Component 3: heterogeneous model voting. Every molecular candidate is screened by three methodologically distinct models—two ML potentials from different families plus a semi-empirical quantum method—whose agreement or disagreement is itself signal. Hard elimination lines (energy above the stability ceiling, non-convergence, HOMO above the oxidation line) are mechanical; disagreement among the three ranks marks a candidate *disputed*:

$$\text{disputed} \iff \text{spread}(r_1, r_2, r_3) \geq \max(2,\ 0.3N), \tag{2}$$

where r_1, r_2, r_3 are the candidate's ranks under the three models among N candidates—forcing reasoning-based refinement rather than acceptance or rejection on a single model's word.

Component 4: the parameter bridge. Microscopic properties are mapped to macroscopic model parameters through a fixed table (Table 1); a mistyped parameter name is rejected by the simulator's validation rather than silently ignored. Every bridge value must carry a source annotation—literature citation or explicit estimate—and estimates are prohibited from entering judgment-relevant outputs unlabeled.

Component 5: mechanized adjudication and the audit chain. Unlike the other components, this one pins the outcome itself. Before the first simulation, the agent writes entry 0 of the audit ledger: the contract criteria, pre-registered in machine-readable form and layered by decision stage, with case-level metadata recording every rule-derived decision and its basis. Thereafter the agent is *forbidden from judging*: every evaluation must be issued by the `log-evaluate` command, which reads the criteria and the tool-output files, compares each metric against its threshold in code, and emits a verdict with an evidence chain—for each checked criterion, the file, the key, the value, and the threshold it was compared against. A design is "achieved" exactly when

$$\text{achieved} = \bigwedge_{c} \left(v_c \geq t_c^{\min}\ \wedge\ v_c \leq t_c^{\max} \right), \tag{3}$$

where the conjunction ranges over the pre-registered criteria of the contract. Derived criteria are computed mechanically; the plating criterion is the min-

Table 1: The parameter bridge: microscopic property to PyBaMM parameter names. Mistyped keys are rejected by the simulator (unknown parameter name), so the table is the safety net against manual transcription error.

Microscopic property	Parameter name (exact key)
Electrolyte diffusivity	`Electrolyte diffusivity [m2.s-1]`
Electrolyte conductivity	`Electrolyte conductivity [S.m-1]`
Cation transference number	`Cation transference number`
Coating, SEI reaction-limited	`SEI kinetic rate constant [m.s-1]`
Coating, SEI migration-limited	`SEI reaction exchange current density [A.m-2]`
Doping, cracking suppression	`Positive/Negative electrode cracking rate`

imum of the anode potential time series against zero, never a threshold the agent may pick:

$$\text{plated} = \left[\min_t \phi_{\text{anode}}(t) < 0\right]. \tag{4}$$

Equations (3)–(4), like every criterion expression in this section, are evaluated verbatim by the adjudication layer. A closing verifier (`verify-deliverables`) checks the deliverable set and, crucially, the audit chain itself: every proposal round must have a same-round mechanical evaluation, and a final judgment entry must exist—a run whose ledger lacks either is refused as incomplete. Judgment is thus an infrastructure layer outside the agent, following the separation of duties in auditing: the entity under audit does not write its own verdicts.

The protocol, the tool library, the adjudication layer, and the cross-harness harnesses are open source (github.com/wangyuanxty/BatteryDesignAgent).

The fifth pipeline stage is the true-compute endorsement: for the closing top candidates only, ab initio DFT (ORCA, r2SCAN-3c [57]) with optional cross-validation in an independent code (CP2K, PBE-D3 [58]), and molecular dynamics for the top electrolyte diffusion coefficient. The endorsement *signs* conclusions; it never screens. In the experiment matrix reported here the endorsement runs are executed for the flagship cases and reported separately (Section 5.8); turning it on or off does not change any contract verdict, by design.

3.3. The Protocol as a Declarative Artifact

The protocol is packaged as a self-contained declarative artifact: a natural-language rule set, a command-line tool library (twenty-plus simulation, evaluation, and audit commands), and the audit ledger format. It binds to no agent framework—any harness that can loop a model over a shell and files can execute it. The governance travels with the contract, not with the harness. In the present study every run executes under one harness; cross-harness portability is a structural property by construction, and its empirical demonstration (the same protocol under additional agent frameworks) is demonstrated empirically with identical contract outcomes (Section 5.7).

3.4. Implementation

The tool library is implemented as a Python package over PyBaMM [4] for electrochemistry—including the thermal-electrochemical parameters [23] and degradation physics [51, 39, 37, 38, 40]—plus surrogate and true-compute runners. Twenty-plus commands cover simulation protocols (discharge at 1C/0.1C/5C, 4C charge at 45 °C, low-temperature discharge, overcharge, 100-cycle aging, nail penetration with a three-reaction thermal-runaway ODE), contract-caliber energy accounting, the mechanical evaluator, and the deliverable verifier. The adjudication layer is pinned by an automated regression suite distributed with the tool library. All conclusion-grade numbers in every report originate in tool outputs; the ledger is the paper's primary data source, and every figure and table here is derivable from it.

4. Experimental Design

4.1. Task Suite

The task suite consists of eight scenario tasks, each a natural-language requirement written as an application scenario plus absolute numeric criteria (Table 2). The scenarios span the design landscape deliberately: a sedan (performance plus thermal safety), grid storage (cycle life plus low temperature), power tools (rate plus power), extreme cold (low temperature plus volumetric density), a flagship vehicle (extreme energy density), a smartphone (volumetric ceiling plus a strict temperature red line plus a voltage plateau), a hybrid vehicle (high-temperature aging plus nail penetration), and a drone (specific energy plus a mass ceiling). Across the eight tasks, fourteen distinct metric types are adjudicated, and all criteria of a task are AND-judged: a design must satisfy every clause of its contract to be "achieved."

Three design choices deserve emphasis. First, the tasks carry no constraint declarations beyond the metrics: nothing states which levers are adjustable, which chemistry to use, or where the baseline sits. Undeclared items are interpreted in the widest sense under protocol rules, with the derivation recorded—this is what forces the agent to make the design-space decisions itself. Second, every number in every task is absolute; no relative-baseline phrasing appears, so an agent cannot redefine the reference it measures against. Third, the tasks are heterogeneous in where their binding constraint lives: T1, T4, and T8 are solvable inside a structural space (thickness, porosity, mass), while T2, T3, T5, T6, and T7 require material, formulation, or transport levers—the distinction on which the comparison with the Bayesian baseline turns.

The contract quantities are defined on standard cell models from the cited literature. Plating is judged on the negative-electrode potential against Li/Li^+, written as the sum of the equilibrium potential and the kinetic overpotential [37, 38]:

$$\phi_a = U_a + \eta_c. \tag{5}$$

The SEI criterion reads a state variable of the degradation model, whose growth law supplies the coating lever that T2 and T6 exploit [39, 51]:

$$\frac{d\delta}{dt} = k_{\mathrm{SEI}}, \tag{6}$$

where k_{SEI} is exactly the parameter the electrode-modification bridge adjusts. The thermal-runaway trigger for the overcharge and nail-penetration criteria is the three-side-reaction energy balance [40]:

$$mc_p \frac{dT}{dt} = \sum_k \Delta H_k r_k(T) - hA\,(T - T_{\mathrm{amb}}), \tag{7}$$

with triggered iff the trajectory crosses the runaway onset; the cooling coefficient h is the thermal-management lever. Criterion values are model-relative—an SEI limit reads the state variable of a specific degradation model, a plateau reads a specific OCP set—so cross-system comparisons are made only within the same model. The scenario anchoring (which production specification each task draws from, and how each threshold was calibrated from the underlying parameter sets) is documented in the supplementary materials; each threshold is the anchoring parameter set's baseline value for that criterion shifted by a fixed increment (e.g., +20% energy density over the baseline configuration, a 5-point cut on 5C retention), never a frontier-tight value—the experiment measures governance contribution at fixed contracts, not frontier performance. The anchoring is a Methods-section explanation and never appears in the task text seen by any agent.

4.2. Conditions and Control Groups

The design follows a treatment-plus-two-controls structure. The governed agent is the treatment; two control groups isolate, respectively, the protocol itself and the classical-optimization boundary. All three conditions share the task texts, the simulation stack, and the session budget, so each observed difference is attributable to the single variable that condition removes.

Table 2: The eight scenario tasks. Every criterion is an absolute number, AND-judged; nothing else is declared in the task text.

ID	Contract (abridged; all values absolute)
T1	Sedan: energy density $\geq$ 392.61 Wh/kg; 4C charge without lithium plating; $T_{\max} \leq 60\,^{\circ}$C; overcharge to 4.7 V without thermal runaway.
T2	Grid storage: energy density $\geq$ 327.18 Wh/kg; 4C charge without plating; SEI $\leq$ 500 nm after 100 cycles; $\geq$ 90% capacity retention at $-20\,^{\circ}$C; SEI $\leq$ 550 nm after 500 cycles.
T3	Power tools: capacity $\geq$ 2 Ah; 5C retention $\geq$ 95%; 4C charge without plating; $T_{\max} \leq 60\,^{\circ}$C; power density $\geq$ 4000 W/kg.
T4	Extreme cold: $\geq$ 95% 1C retention at $-20\,^{\circ}$C; energy density $\geq$ 327.18 Wh/kg; volumetric energy density $\geq$ 880 Wh/L.
T5	Flagship: energy density $\geq$ 500.94 Wh/kg; 4C charge without plating; $T_{\max} \leq 60\,^{\circ}$C.
T6	Smartphone: volumetric energy density $\geq$ 950 Wh/L; 4C charge without plating; $T_{\max} \leq 50\,^{\circ}$C; SEI $\leq$ 500 nm after 100 cycles; voltage plateau $\geq$ 4.1 V.
T7	Hybrid: energy density $\geq$ 327.18 Wh/kg; 4C charge without plating; SEI $\leq$ 550 nm after 100 cycles at $45\,^{\circ}$C; nail penetration (10 W short-circuit heat) without thermal runaway.
T8	Drone: energy density $\geq$ 446.18 Wh/kg; 5C retention $\geq$ 90%; cell mass $\leq$ 40 g.

Governed agent (G). The full protocol of Section 3, executed under two large language models separately—deepseek-v4-pro and deepseek-v4-flash—giving eight runs per model and a model-robustness axis. Two further legs re-execute the eight sessions under GPT-5.6 Luna and under mimo-v2.5—foreign-vendor models served through OpenAI-compatible APIs, same protocol, harness, task texts, and budget—so the cross-family comparison is made outside the primary vendor at matched tier logic.

Protocol-free agent (C1). The same LLM, the same agent harness (tool-use loop), the same tools available in the environment (a Python interpreter with the simulation libraries, shell, and file tools), and the same session budget—but no protocol: the system prompt carries only a role statement, the tool descriptions, a ban on true-compute calls, the budget, and the deliverable request. There are no stage rules, no funnel, no pre-registered criteria, no mechanical evaluator, and no audit-chain verifier. C1 is the *protocol-removal control*: it isolates the governance layer by holding every other resource constant—model, harness, toolset, budget, task texts. One run per task, deepseek-v4-pro.

Bayesian optimization (C2). A classical BO baseline in the architecture parameter space (ten structural parameters: electrode thicknesses and porosities, separator thickness and porosity, current collector thicknesses, heat transfer coefficient, negative particle radius), searching the objective *energy density minus safety penalties*,

$$\mathrm{obj} = \mathrm{ED} - \sum_k \lambda_k \nVdash [v_k \not\succeq t_k], \tag{8}$$

with a fixed penalty set that covers every contract criterion except the T1 overcharge-runaway trigger (plating, temperature limit, SEI, plateau, mass); a criterion absent from this set is invisible to the search. The material-bottleneck failures are therefore not a visibility artifact: SEI and plateau *are* penalized (a re-run with the aligned $50\,^{\circ}\mathrm{C}$ red line confirms the failures persist, Section 5.1). What the optimizer cannot do is move a chemistry switch, because its knob space is structural only—and that boundary is the point: the configuration replicates Battery-Sim-Agent's published BO standard [5]: Meta's Ax platform, Sobol initialization, GPEI with LogNEI acquisition, fixed seed 1234, fifty evaluations per task—the same simulator and the same contract-caliber energy accounting as the agents, so the comparison is on identical physics. The search is deterministic in its seed; a three-seed replication (1234, 5678, 9012) reproduces the attained-task set $\{T1, T4, T8\}$ identically, with per-task best objectives within 9% across seeds (supplementary materials). C2 is the *classical-baseline control*: it isolates the solution-space boundary of ungoverned search—a strong, published-standard optimizer that cannot leave its parameter space. One run (fifty evaluations) per task.

Resources held constant. All three conditions receive the same task texts, the same simulation stack, and the same session budget: the agents run under the same harness turn budget—the agent SDK's `max_turns`=300 setting—and the protocol carries no round cap of its own. Table 4 reports a log-derived activity metric, *assistant messages*, which is finer-grained than the harness's internal turn counter; the budget *setting*, not the message count, is the controlled resource. The two units are empirically anchored: the one session that hit the budget (the protocol-free agent on T6) did so at 443 assistant messages. Every other session concluded before the cap. The BO baseline's fifty evaluations define its budget. A second, simulator-native currency is likewise anchored: the governed sessions average 25 run-pyamm calls per task (range 0–55; 31 including calc-energy), against BO's fixed fifty—the agents hold no evaluation-count advantage, and both conditions sit in the same order of magnitude on every resource axis. Neither currency is equalized per operation; each condition receives its method family's standard budget, and the comparison is made at that level. True-compute endorsement is disabled across the main matrix (`real_compute` = `false`); endorsement results are reported separately in Section 5.8.

4.3. Ablation Design

The governance components are fixed properties of the protocol; ablation is an experimental intervention that relaxes exactly one component per run. An ablation variant is declared in the task text itself ("ceiling escalation disabled")—keeping the natural-language requirement the sole entry point—and the declaration is recorded in the run's entry-0 metadata. Three ablations are executed: *ceiling escalation off* (no proactive material-design escalation; the agent tunes within the current system unless the text names materials), on T2/T5/T6; *architecture forcing off* (no required 2–4 architecture variants per round), on T1/T2/T3/T4/T7/T8; and *funnel voting off* (molecular screening by a single ML potential instead of three-model heterogeneous voting), on T2/T5/T6. Each ablation run uses deepseek-v4-pro and the full protocol otherwise. Twelve ablation runs in total.

4.4. Judgment Machinery

All verdicts reported in this paper come from the mechanized adjudication layer of Section 3: criteria pre-registered in entry 0, verdicts computed by `log-evaluate` from tool-output files, evidence chains per criterion, and the deliverable verifier's audit-chain checks. Two further audits are applied to the control conditions. For C1, the agent's own "achieved" claims are re-adjudicated at the protocol's contract caliber by the authors: every claimed metric is recomputed mechanically from the artifact files the agent produced, on the protocol's own definitions. Two calibers recur throughout and are fixed here:

$$\mathrm{ED} = \frac{\int V(t)\, I\, dt}{\sum_i l_i\,(1-\varepsilon_i)\,\rho_i \cdot A}, \tag{9}$$

the contract-caliber energy density (mass = layer stack, electrolyte excluded), and

$$\mathrm{ret}_{5C} = C_{5C}/C_{1C}, \tag{10}$$

the rate-retention caliber against the 1C reference. The power criterion is computed from the discharge-start voltage drop and the cell mass (standard definitions of DC resistance and peak power):

$$\mathrm{DCR} = \frac{V(0) - V(t_{10\%})}{I}, \qquad P_{\max} = \frac{V_{\mathrm{oc}}^2}{4\,\mathrm{DCR}\cdot m}, \tag{11}$$

with $V(t_{10\%})$ the voltage at 10% of discharge time. A claim computed against a different reference (e.g., 5C over C/3) or a different mass model is a different caliber and is recomputed (nail penetration means a thermal-runaway trigger criterion, not a steady-state temperature). For C2, the objective trajectory is recomputed from the logged evaluation outputs. The re-adjudication rules are mechanical and are listed in the supplementary materials.

4.5. Real-Data Anchor

To anchor the simulation world to a physical artifact, the calibrated parameter set for the LG M50 cell [23] is compared against the beginning-of-life 0.1C discharge of a production LG M50T 21700 cell from the Imperial College degradation dataset [21, 22]: measured capacity 4878.9 mAh versus simulated 4857.0 mAh (+0.45%), and voltage-curve RMSE 20.6 mV over 2–82% depth of discharge. The alignment is the Methods-section calibration record; it establishes that the numbers agents are judged against are anchored to a production cell—a validation of the simulator, not of the final designs (Section 6). Figure 6 overlays the measured and simulated curves.

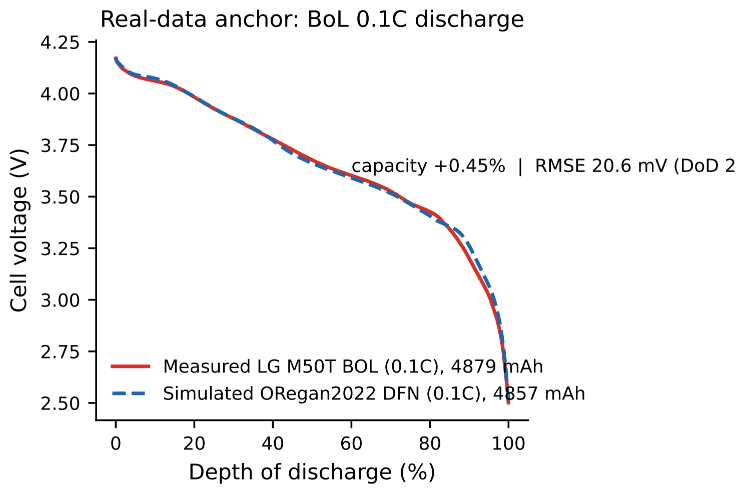


Figure 6: Real-data anchor: beginning-of-life 0.1C discharge of a production LG M50T 21700 cell (Imperial College dataset, [21]) versus the ORegan2022-parameterized DFN simulation [23]. Capacity agrees to +0.45%; voltage RMSE 20.6 mV over 2–82% depth of discharge.

4.6. Cross-Harness Portability

The protocol's independence from any specific agent framework (Section 3.4) is a design property of the artifact itself; its empirical demonstration is reported here and executed in Section 5.7, so that its results slot into Section 5 without methodological ambiguity. The design: the identical protocol—the same declarative rule set, the same command-line tool library, and therefore the same adjudication layer—runs the same task texts (T1, a structural-bottleneck contract, and T6, a material-bottleneck contract) under the same LLM and the same nominal budget setting (300, in each harness's own turn unit) across three harnesses: the harness used throughout this paper, the OpenAI Agents SDK, and LangChain. The measured quantities are contract attainment (mechanically judged, identical criteria) and audit-chain completeness (the verifier's checks). Because the adjudication layer is a command-line tool rather than harness code, the verdict mechanics are harness-independent by design; the experiment measures whether agent behavior under the same governance remains contract-faithful when the orchestration substrate changes, as reported in Section 5.7.

5. Results

5.1. Main Comparison: Contract Attainment

Table 3 reports contract attainment for all eight tasks under the six columns, and Figure 7 shows the same matrix as a color-coded attainment map. Three findings stand out. First, the governed agent achieves **8/8** under the two primary models and **4/8** under each of the two foreign-vendor models (Section 5.4); every verdict is mechanical, with an evidence chain per criterion and a verifier-passed audit chain. Second, the Bayesian optimizer achieves **3/8**—exactly T1, T4, and T8, the three tasks whose binding constraint is structural (thickness, porosity, mass). Its best designs reach energy densities above the governed agent's on these tasks (e.g., 678 versus 472 Wh/kg on T4), the expected consequence of an unconstrained thin-electrode search against a margin-balanced design; on the remaining five tasks it fails on the contract's own criteria—4C plating in T2, T3, T5, T6, T7; SEI limits in T2, T6, T7; a catastrophic 3.44% 5C retention on T3; the 4.1 V plateau unreachable on T6, where the best design stalls at the NMC811 plateau limit (4.0065 V)—a re-run with the penalizer aligned to the contract's 50 °C red line (the main run hardcoded 60 °C) leaves the outcome unchanged: 0/50 trials attain the contract in either configuration, 30/50 clear the volumetric criterion alone in the aligned run, and none is simultaneously plating-free and below the red line; nail penetration triggered on T7. Figure 8 shows the BO trajectories: the three structural tasks climb to attainment, the five material-bottleneck tasks plateau against penalties no structural parameter can remove. Third, the protocol-free agent achieves **none** within the measurement space: two of its claimed successes (T4, T8) solve the contract by switching to a lithium-metal cell outside the validated parameter sets, and its remaining claims fail mechanical re-adjudication (Section 5.5). The tie between the two baselines (three achieved tasks each) with orthogonal failure modes is the paper's central observation: the bare agent and the black-box optimizer fail for different, specifiable reasons, and only the governed agent passes on all eight contracts.

Contract-margin calibration deserves explicit reporting, since "8/8" could otherwise read as loose thresholds. The thresholds are fixed increments over the anchoring parameter sets' baseline values, not frontier-tight values (Section 4.1), and all three conditions face identical contracts—any looseness benefits the baselines equally. The achieved margins show the contracts binding where they are meant to bind: on six of the eight governed runs the last criterion to clear sits within 4.5% of its threshold (T6's plateau at +0.4% and thermal line at +0.5%; T8's mass at +0.5%; T3's and T5's thermal lines at +1.0–1.5%); the two exceptions are T2's low-temperature retention (+10.6%) and T7's SEI limit (+15.4%). The wide margins concentrate on the energy-density clauses of the structural tasks (+41–48%), where an unconstrained thin-electrode search overshoots cheaply; the same mechanism behind the BO's 678 versus 472 Wh/kg on T4, and precisely the margin the governed designs partially spend on the safety and durability criteria. Per-criterion margins for all achieved runs are tabulated in the supplementary materials. The SEI criteria deserve one further disclosure, because they are the single criterion class whose bridge lever (the electrode-modification coating scaling the SEI kinetic rate constant) can move a verdict: T2 shipped a $10^{-4}\times$ rate constant (declared in the audit as an estimate beyond the bare-ALD literature magnitude), yet the identical design passes the contract at $10^{-2}\times$ (500.4 nm against the 550 nm limit, a 9% margin); T6's SEI work is not verdict-bearing; T7 used no kinetic override (supplementary materials).

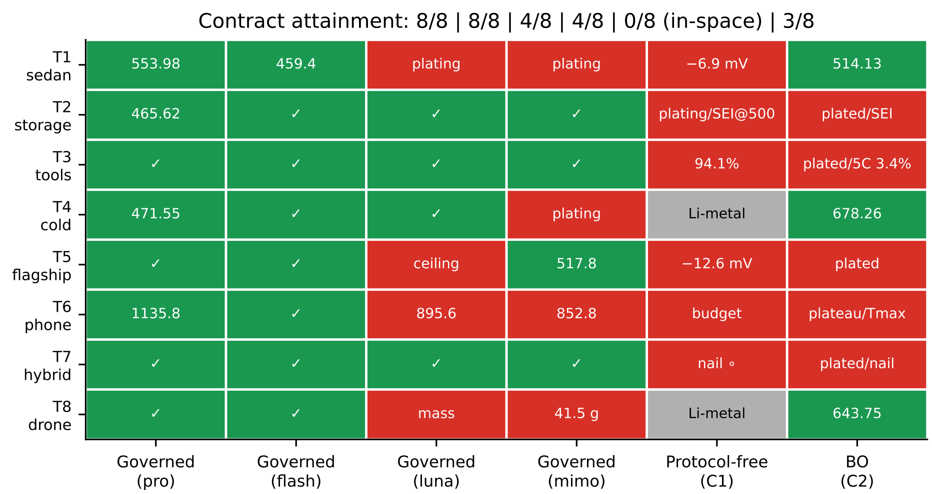


Figure 7: Contract attainment across the eight tasks and four conditions. Green: achieved under mechanical verdict; red: contract criteria failed; gray: outside the measurement space (C1 on T4/T8 switched to a lithium-metal cell outside the validated parameter sets). Numbers are key re-adjudicated values (energy density in Wh/kg on T1/T4/T6, Wh/L basis for T6 shown as 1135.8; margins in mV; SEI in nm; 5C retention in %).

5.2. Resource Cost

Table 4 reports the governed agent's resource consumption per task under both models—assistant messages (activity measure) and model tokens

Table 3: Contract attainment, eight tasks × six columns (governed agent shown under four models: pro, flash, luna, mimo). ✓ = achieved (mechanical verdict); × = failed contract criteria; ∘ = outside the measurement space (see text). G = governed agent; C1 = protocol-free agent; C2 = Bayesian optimization. Key re-adjudicated numbers in parentheses. Per-task numbers are r1 values; the condition totals are seed-robust as detailed in the table note and supplementary materials.

Task	G (pro)	G (flash)	G (luna)	G (mimo)	C1	C2
T1 sedan	✓ 553.98	✓ 459.4	× (plating)	× (plating)	× (−6.9 mV)	✓ 514.13
T2 storage	✓ 465.62	✓	✓	✓	× (plating, SEI)	× (plating, SEI)
T3 tools	✓	✓	✓	✓	× (5C 84.8%)	× (plating, 5C)
T4 cold	✓ 471.55	✓	✓	× (plating)	∘ (Li-metal)	✓ 678.26
T5 flagship	✓	✓	× (ceiling)	✓ 517.8	× (−12.6 mV)	× (plating)
T6 phone	✓ 1135.8	✓	× (895.6 Wh/L)	× (852.8 Wh/L)	× (budget)	× (plateau, $T_{\max}$)
T7 hybrid	✓	✓	✓	✓	× (nail unproven)	× (plating, nail)
T8 drone	✓	✓	× (mass)	× (mass, $T_{\max}$)	∘ (Li-metal)	✓ 643.75
Total	**8/8**	**8/8**	**4/8**	**4/8**	**0/8 in-space**	**3/8**

G (pro): three-seed replication (r1–r3); every attained task achieved in all three seeds (24/24, supplementary materials). G (flash), G (luna), G (mimo), and C1: one run per cell, as stated in the respective sections. C2: three seeds (1234/5678/9012); attained set $\{T1, T4, T8\}$ in all three. The mimo T1 run omitted the final ledger entry; under the audit-chain rule it is counted as not attained (Section 5.4).

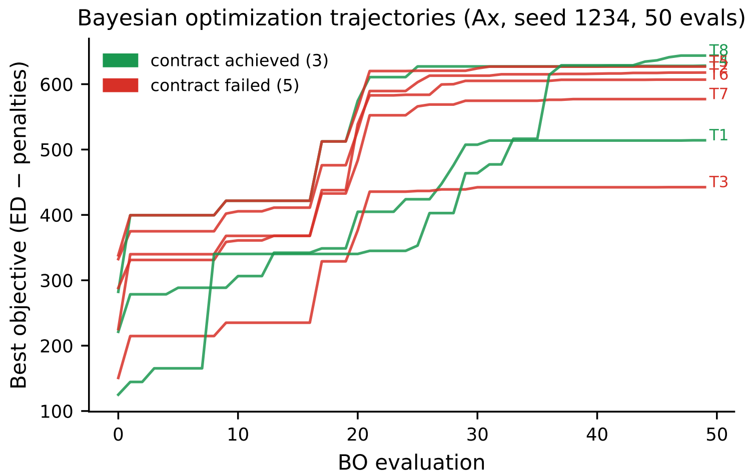


Figure 8: Bayesian optimization best-objective trajectories per task (Ax, Sobol initialization, GPEI + LogNEI, seed 1234, 50 evaluations). The three structural tasks (T1/T4/T8) achieve their contracts; the five material-bottleneck tasks plateau against penalties (plating, SEI, plateau, nail) that no structural parameter can remove.

(input/output)—and Figure 9 plots the same data, highlighting the material-bottleneck peaks (T6 and, under flash, T5). Three observations. Material-bottleneck tasks are the expensive ones: T6 (the smartphone contract demanding a cathode-system switch) consumes 468 messages and the largest token counts under pro, and T5/T6 dominate under flash. The economy model completes the suite at roughly three-quarters of the flagship model's input tokens (1.12M versus 1.52M) and 2,418 versus 2,723 assistant messages, with zero attainment difference: the protocol, not the model, absorbs the difficulty. The C1 runs run under the same harness budget setting per task (one, T6, hit it); the BO baseline consumes fifty simulator evaluations per task (400 in total).

Table 4: Governed-agent activity per task (assistant messages; tokens in/out in thousands). Assistant messages are a log-derived activity measure; the controlled budget is the identical harness setting across all sessions.

	pro		flash	
Task	msg.	tok. in/out	msg.	tok. in/out
T1	303	232/176	258	128/75
T2	380	210/202	191	82/83
T3	263	173/136	244	136/91
T4	327	196/139	323	152/129
T5	350	164/169	376	149/141
T6	468	235/234	437	207/245
T7	371	149/148	374	175/142
T8	261	158/147	215	94/87
Total	2723	1517/1352	2418	1123/992

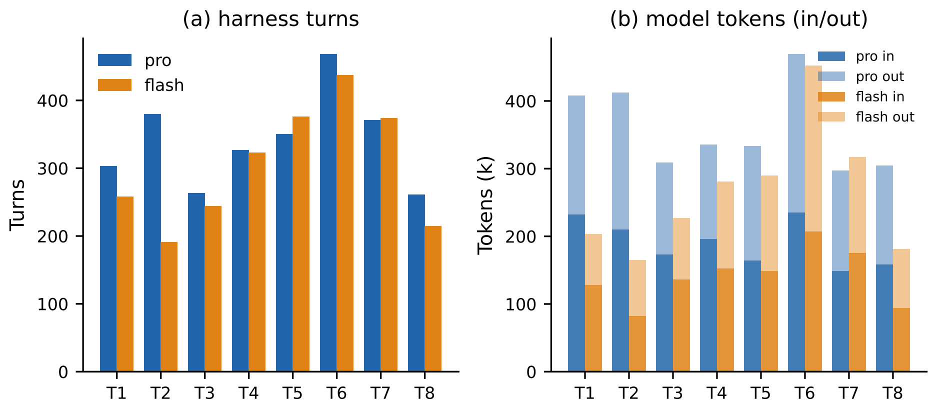


Figure 9: Governed-agent resource cost per task under both models: (a) assistant messages; (b) model tokens in/out (thousands). Material-bottleneck tasks (T6, and T5 under flash) are the expensive ones; the economy model completes the suite at 8/8 with roughly three-quarters of the flagship model's input tokens.

5.3. Ablation: Which Components Carry the Effect

Table 5 reports the twelve ablation runs, and Figure 10 summarizes their verdicts as a per-component stacked bar. The three mechanisms contribute at very different magnitudes, and the decomposition is the paper's answer to RQ2. Ceiling escalation is the load-bearing mechanism: disabling it costs two of the three material-bottleneck tasks outright—T6 loses its LNMO system switch (the 4.1 V plateau is unreachable in NMC811 space) and T2 loses its SEI-kinetics bridge (three of five criteria pass, the in-space result)—while T5, whose tipping point turned out to live inside the configuration space, is unaffected. Architecture forcing is a narrow lever: disabling it changes exactly one task (T2), where forced breadth was the exploration that found the design. Funnel voting shows no verdict change in any of its three cells: with the three-model vote off, the molecular path was actually taken *more* often (six molecules screened on T2, three on T6, versus none in the baseline runs), and the single-model screenings still led to achieved contracts. In this task suite the voting mechanism showed no verdict change in any cell—one run per cell, so the null is an observed absence rather than a statistical claim; it is consistent with the mechanism's role as defensive infrastructure that the baseline never exercised, and with Battery-Sim-Agent's reported low-headroom calibration [5].

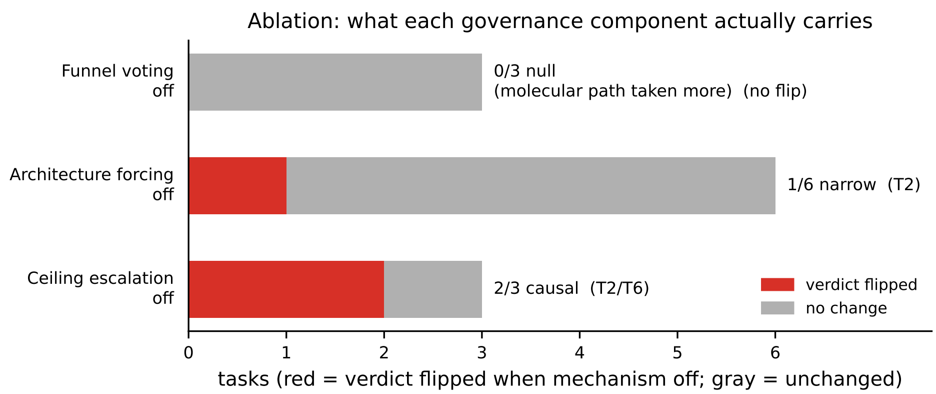


Figure 10: What each governance component actually carries: red segments count the tasks whose verdict flipped when the component was disabled; gray segments count the unchanged tasks. Ceiling escalation is causal for two of three material-bottleneck tasks; architecture forcing for one of six; funnel voting for none—no verdict change observed in this suite.

5.4. Model Robustness

The governance is model-robust: 8/8 under the two DeepSeek models, with the same contracts attained through different designs—and under the pro model the 8/8 is not an artifact of a single run: all eight tasks were repeated under three seeds, and every attained task achieved in all three (24/24, supplementary materials). On T5 the pro run settled on a dual-validated architecture design while the flash run took the material-funnel

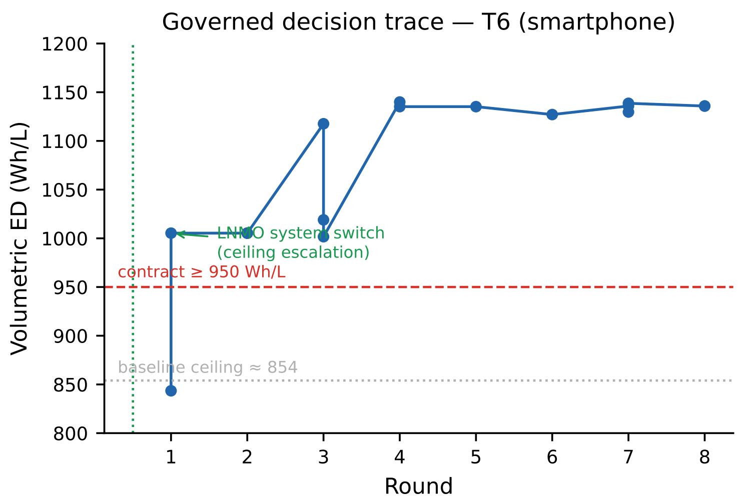


Figure 11: Governed decision trace on T6 (smartphone contract): volumetric energy density by round, with the contract line (950 Wh/L) and the baseline ceiling (854 Wh/L). The jump across the dashed line is the ceiling escalation—the LNMO system switch—followed by rounds of refinement to 1135.8 Wh/L.

Table 5: Ablation results: verdict with the component disabled versus the full-protocol verdict (F). ✓ = achieved; × = not achieved. The T5 voting-off cell is a no-op: the molecular funnel was not exercised in that task (start stage = cell design), so the cell carries no inference about the voting mechanism.

Task	ceiling off	force off	voting off	full protocol
T1	–	✓	–	✓
T2	× (3/5)	×	✓	✓
T3	–	✓	–	✓
T4	–	✓	–	✓
T5	✓	–	✓	✓
T6	×	–	✓	✓
T7	–	✓	–	✓
T8	–	✓	–	✓

route (LNMO, with FEC/VC/PES/DTD passing the funnel); on T1 the designs differ by 94 Wh/kg in opposite-margin strategies yet both meet every criterion. Contract attainment is reproducible across models; the solution space remains broad. One execution defect surfaced and was absorbed by the audit layer: the flash T6 run delivered a verifier-passed deliverable set but omitted the final judgment entry from its ledger; the mimo T1 run repeated the same omission. The verifier now refuses such incomplete audit chains and counts the run as not attained—the defect became a regression test, which is precisely the purpose of the audit infrastructure.

Two further legs extend the robustness question across vendor families: the same eight sessions re-executed under GPT-5.6 Luna and under mimo-v2.5—foreign-vendor models served through OpenAI-compatible APIs—with the same protocol, harness, task texts, and budget. Both legs attain 4/8—Luna on T2, T3, T4, T7 and mimo on T2, T3, T5, T7—and their non-attainments differ revealingly from the baselines'. First, both are honest: Luna's four are mechanical *not-achieved* verdicts with complete evidence chains, and mimo reports its failing criteria in the recommendation itself (T8 is self-labeled "achieved with documented limitations" because mass and thermal fall outside the contract; T6 is a mechanical not-achieved). Second, the intersection $\{T2, T3, T7\}$ is attained under every one of the four models—the stability point of the governance claim—while the disagreements are model-specific and informative: T4's plating line survived Luna's design but not mimo's, whose final design report concedes the plating clause even as its ledger carries "achieved" (an overdeclaration the audit records as such); on T5, mimo found the thin-electrode plus high-transport-electrolyte route that eluded Luna. Third, the joint non-attainments concentrate exactly where the governed DeepSeek runs needed the ceiling-escalation mechanism: T1 falls to plating under both foreign-vendor legs, T6 stalls at the NMC811 plateau (mimo: 852.8 Wh/L and 3.97 V), and T8 stalls at the mass boundary (41.5 g) plus its thermal line (361.8 K against the protocol's 333.15 K safety default). The protocol supplied the machinery; the models' searches did not reach the material lever, so the honest result is a lower score rather than a shifted criterion—the same failure the ablation already pointed to, observed now from the model side. One yardstick observation deserves recording: mimo's T8 run pre-registered the thermal threshold at 358.15 K against the protocol's 333.15 K safety default—a standard shift on the agent's own side—yet its 361.8 K exceeds even the shifted standard, so the verdict is unchanged; the audit chain keeps both registered standards visible. Governance transfers across model families; capability does not. With 4/8, each foreign-vendor leg still clears the classical baseline's 3/8, and on tasks the optimizer cannot touch at all (T2, T3, T7). Like every other condition column, both legs are one run per cell: their reading is at the level of the failure *pattern* (honest, distinct from the baselines') rather than per-task inference. Figure 12 shows the four model legs side by side: the same governed loop attains the same contracts through different designs (and, on T6, identifies the same LNMO bottleneck).

5.5. Yardstick Evidence: The C1 Re-Adjudication

Table 6 shows what the protocol-free agent's "achieved" claims contain when re-adjudicated at the contract caliber. The failure modes map one-to-one onto the yardstick failure anticipated in Section 1: a shifted threshold (T1: -6.9 mV declared safe against a self-set -10 mV dendrite onset, where the contract is zero tolerance); an undeclared model substitution (T2: its 8.1/15.1 nm SEI claims came from PyBaMM's solvent-diffusion-

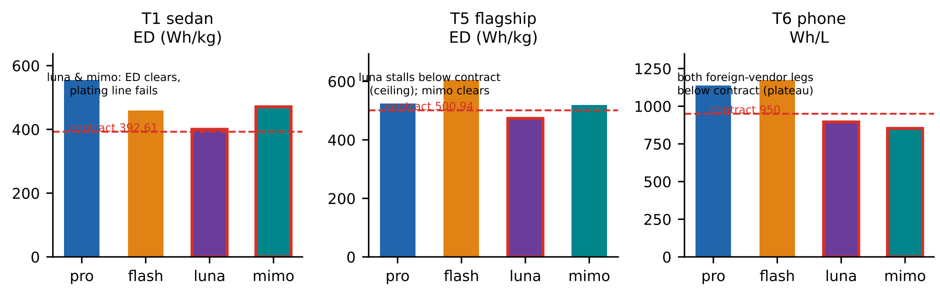


Figure 12: Model robustness with design diversity: the same contracts attained under the four models with different designs (T1: 553.98 vs 459.4 Wh/kg and a plating failure on both foreign-vendor legs; T5: 524.2 vs 603.6 vs 517.8). Contract attainment is reproducible; the solution space stays broad.

limited growth law rather than the contract's ec-reaction-limited model; re-adjudicated at the contract caliber with the full declared design, the 500-cycle SEI is 818.9 nm against the 550 nm limit, and the 4C charge plates at −360.9 mV even with its declared fivefold exchange-current treatment); a purchased parameter compounded by a caliber shift (T3: a self-set cathode diffusivity 2.5× the standard value, and "96.7% retention" measured against C/3, where the contract's 5C/1C caliber gives 94.1% with its declared parameters and 84.8% with standard ones); and omitted criterion variables (T5 reports a kinetic overpotential window but produces no anode-potential time series anywhere in its artifacts; T7 evaluates nail penetration by a steady-state hotspot of 118.5 °C with no trigger criterion; T6 hit the harness budget—the only session in the study to do so—without delivering a design). The two remaining claims (T4, T8) achieve the contract by leaving the measurement space—a lithium-metal chemistry outside the validated parameter sets—a legal reading of the task text, but outside the protocol's verification chain, and therefore not counted. The comparison is not “the bare agent is incompetent”: it is that an ungoverned agent’s measurement standard is movable, and only the adjudication layer pins it. Figure 13 walks through the four mechanisms by which the standard was moved, against the contract values.

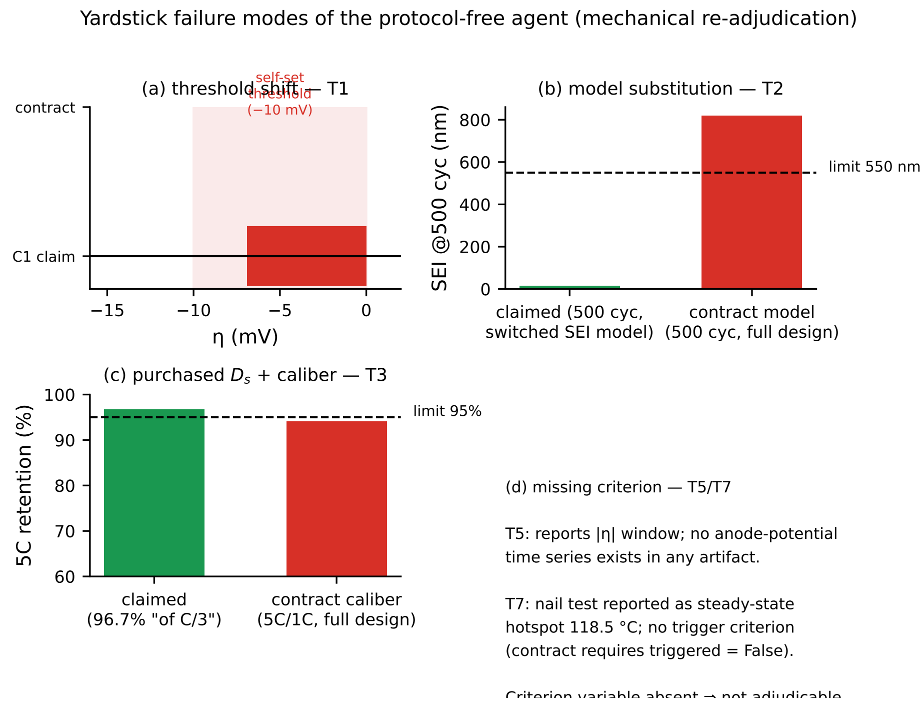


Figure 13: The yardstick failure modes of the protocol-free agent, re-adjudicated at the contract caliber: (a) threshold shift—T1’s −6.9 mV declared safe against a self-set −10 mV onset where the contract is zero tolerance; (b) model substitution—T2’s SEI claimed at 15.1 nm (500 cycles) under the solvent-diffusion-limited growth law, 818.9 nm under the contract’s ec-reaction-limited model with the full declared design; (c) purchased parameter plus caliber shift—T3’s 96.7% retention against C/3 versus 94.1% at the contractual 5C/1C with its declared diffusivities (84.8% with standard ones); (d) missing criterion variables—T5’s absent anode-potential series and T7’s nail test without a trigger criterion.

Table 6: C1 (protocol-free agent) claims versus contract-caliber re-adjudication. All re-adjudications mechanical (definitions in supplementary materials).

Task	Self-reported	Re-adjudicated	Yardstick failure
T1	"safe" plating margin	× (-6.9 mV < 0)	self-set -10 mV threshold
T2	SEI 8.1/15.1 nm, "all verified"	× (plating -360.9 mV; SEI 818.9 nm @500 cyc)	undeclared SEI model substitution
T3	5C retention 96.7% (of C/3)	× (94.1% 5C/1C)	purchased D_s (2.5×) + caliber shift
T4	achieved	◦	out-of-space (Li-metal)
T5	$\lvert\eta\rvert$ "3–4× safe"	× (-12.6 mV)	missing criterion variable
T6	(no deliverable)	×	budget exhausted
T7	"all met"	×	missing trigger criterion
T8	achieved	◦	out-of-space (Li-metal)

5.6. Case Study: The T6 Contract, End to End

Aggregate attainment says the governed agent passes; a walk through one case shows what the machinery actually produces. We pick T6—the smartphone contract, the task on which *both* baselines failed (BO against the 4.1 V plateau dead-line, C1 against its turn budget)—and replay the run from its artifacts: the audit ledger, the auto-generated report, and the deliverable package.

Pre-registration. The agent's first act is entry 0 of the ledger: the contract parsed from the task text into machine-readable criteria, with every rule-derived decision and its basis recorded (Table 7 shows the operative rows). The starting-stage decision is explicit: the task names no system, so the anchor table maps to the Chen2020 default; the plateau clause is flagged from the outset as the suspected material bottleneck. The run's full documentation is a deterministic seven-section rendering of the ledger (overview, trajectory, funnel, stage results, endorsement, final recommendation, notes)—a rendering of the audit chain, not a narrative written after the fact.

Ceiling assessment and escalation. The Stage-1 plan opens with a baseline characterization whose recorded conclusion is the ceiling assessment in action: "R1 — baseline characterization (ceiling assessment): Chen2020 default 1C discharge (evidence: plateau infeasibility $\rightarrow$ material bottleneck)". The agent then proposes the system candidate `systemLNMO` (a 4.7 V-class spinel parameter set), which per the candidate-type rules bypasses the molecular funnel—system switches are screened at the cell scale, not by ML potentials—and the funnel entry records this routing explicitly. (The LNMO parameter set is an author-provided 4.7 V-class spinel set derived from the literature; its provenance is recorded in the run's entry-0 meta-

Table 7: Entry-0 criteria as the agent pre-registered them for T6 (excerpt; the full entry also records the five-category freedom parsing, the real-compute setting, and the start-stage basis).

Layer	Metric	Threshold (as parsed)
stage2 (cell)	volumetric ED	≥ 950 Wh/L
stage2 (cell)	voltage plateau	≥ 4.1 V (midpoint)
stage2 (cell)	SEI @ 100 cyc	≤ 500 nm
stage3 (safety)	4C max temperature	≤ 323.15 K ($50\,^{\circ}$C, mechanical conversion)
stage3 (safety)	plating	false

data.) The escalation is a logged, reasoned decision, not a silent pivot.

Iteration. Table 9 shows the round trail from the mechanical evaluations (the full trajectory is plotted in Figure 11). The first LNMO architecture (R02, 1005.3 Wh/L) already clears the volumetric criterion but *fails* other criteria; rounds R03–R06 refine thickness and formulation until R07 passes all criteria, with R08 confirming. The trail is a genuine fail–fail–$\cdots$–pass sequence—the ledger records the failures, which is what makes the final pass evidence rather than assertion.

Table 8 condenses the run's full candidate sheet—twenty proposals, each logged with its name, type, and one-line rationale. The sequence reads like a design engineer's notebook: thermal probes first, then plating fixes on three distinct levers (anode capacity headroom, anode kinetics, electrolyte transport), then SEI levers, a combined candidate, DFN confirmation, and finally DFN-gap fixes with a margin-widening probe. Every pivot is a recorded decision with its hypothesis attached.

Table 8: The T6 candidate sheet (condensed from the report's full list of twenty; each proposal is logged with its rationale in the ledger).

Round	Candidate	Purpose (logged rationale)
R01	systemLNMO	system switch: 4.7 V-class LNMO spinel
R01	baselineChen2020	anchor-table default baseline
R02	lnmo_h200 / h500	thermal probes at 4C ($h = 200/500$)
R03	np120 / nr35 / elyte	plating fixes: N/P headroom; anode kinetics; electrolyte transport
R03	sei6e13	SEI-suppression probe (electrode modification)
R04	np125 / np120_nr35	plating closure: N/P 1.25 and a leaner alternative
R04	sei_c2270_d12 / d10	SEI levers: EC concentration and D_{EC} cuts
R05	lnmo_final	combined final candidate
R06	lnmo_final	DFN confirmation of the combined candidate
R07	r7_h400_d5 / nr30 / combo	DFN-gap fixes: thermal; plating; combined
R08	r8_final / r8_d6_margin	final candidate and margin-widening probe

Table 9: T6 round trail: one row per design round, reconstructed mechanically from the run ledger. “Agent reasoning” gives a verbatim thinking excerpt from the run log where one exists (shown in italics), followed by the decision rationale the agent itself recorded at that round; “Simulation artifacts” lists the evidence files the mechanical evaluator read; “Key values” are the decisive contract-caliber numbers.

Round	Agent reasoning (recorded rationale)	Simulation artifacts	Key values	Verdict
Entry-0	Task text names a smartphone use case; the plateau $\geq$ 4.1 V clause is flagged from the outset as the suspected material bottleneck (NMC811-class midpoint $\approx$ 3.6 V).	parameter-set anchor check	criteria pre-registered, five freedom classes	contract locked
R01	Baseline vs. system switch: the Chen2020 baseline cannot reach the plateau; LNMO at defaults meets ED and midpoint.	r1_chen2020_1c, r1_lnmo_1c, calc-energy $\times$2	ED_l 843.5 / 1005.3; Vmid 4.166	$\times$ base / ✓ LNMO
Funnel	Molecular funnel over the screening set; candidates benchmarked by three-model vote.	run-mlp (MACE, CHGNet), run-xtb	four molecules below the HOMO line	funnel passed
R02	Gap quantification on the LNMO design: plating and SEI >500 nm; SEI growth is temperature-insensitive, so cooling is not the lever.	r2_lnmo_aging; r2_lnmo_4c (h=200/500)	plated; SEI 753.1	$\times$ $\times$ $\times$

Round	Agent reasoning (recorded rationale)	Simulation artifacts	Key values	Verdict
R03	*"Round 3 results: np120 (N/P 1.2) 321.20 K,* −0.0049 *V—STILL plated (borderline!)"* (thinking): N/P dominates plating (end-of-charge kinetics); particle radius weak; electrolyte transport direction dropped.	r3_np120_4c, r3_nr35_4c, r3_elyte_4c	SEI 736.0 nm (aging probe)	× × ××
R04	SEI probes from the rate law: c_0 and D_{EC} cuts (0.5×) bring SEI below 500 nm.	r4_np125, r4_np120nr35, r4_sei1/2_aging	SEI 395.2 nm	×/✓✓✓
R05	Combine the measured levers into one shipping candidate; run the complete SPMe evidence battery.	r5_final__spme (1c/aging/4c)	1135.2 Wh/L; 4.188 V; 321.09 K; SEI 435.4	✓ (SPMe)
R06	*"DFN confirmation FLIPPED two criteria: T_max 323.80 K; plated TRUE"* (thinking): proxy layering rule—conclusion-grade numbers must come from DFN; the SPMe win is re-checked at high fidelity.	r6_final__dfn (energy/aging/4c)	1127.1 Wh/L; Vmid 4.097; plated	×

Round	Agent reasoning (recorded rationale)	Simulation artifacts	Key values	Verdict
R07	DFN flipped three criteria; probe the three fixes in isolation and in combination (h=400, r=3.5, D_e=5e-10).	r7_d5, r7_nr30, r7_combo (DFN)	tip 1138.6 Wh/L; 321.62 K	✓/×/×
R08	*"Round 8 evaluated: both candidates pass with all 5 criteria checked"* (thinking): final combined candidate through the full DFN chain; deliverable package verified.	r8_d6__dfn (energy/aging/4c)	1135.8 Wh/L; 4.1145 V; 321.42 K; SEI 385.3	✓✓
Endorsement	ORCA (r2SCAN-3c) plus CP2K cross-validation of the funnel finalists; all four below the −6.0 eV line in both codes.	run-orca, run-cp2k (Fig. 15)	endorses screening	signed
Final	Ledger closed: recommendation recorded, deliverables and index generated, verifier pass.	`verify-deliverables`	VBF-T6R1-* package	achieved

Mechanical adjudication. The closing evaluation's evidence chain is computed by the adjudication layer, criterion by criterion, each pointing at the exact file and key that supports it: `cell/r8_d6_energy_dfn.json:energy_density_wh_l` = 1135.83 vs. threshold ≥ 950 (pass); `cell/r8_d6_energy_dfn.json:midpoint_voltage_v` = 4.1145 vs. ≥ 4.1 (pass); `cell/r8_d6_aging_dfn.json:sei_thickness_nm_end` = 385.29 vs. ≤ 500 (pass); T_{max} 321.42 K vs. ≤ 323.15 K (pass); plating false (pass). Every "achieved" in this paper is backed by such a chain. Figure 14 shows the two decisive physics behind the case: the plateau gap that forced the system switch, and the plating margin the criterion judges.

Deliverable package. The run closes with the seven-file industry-standard package—

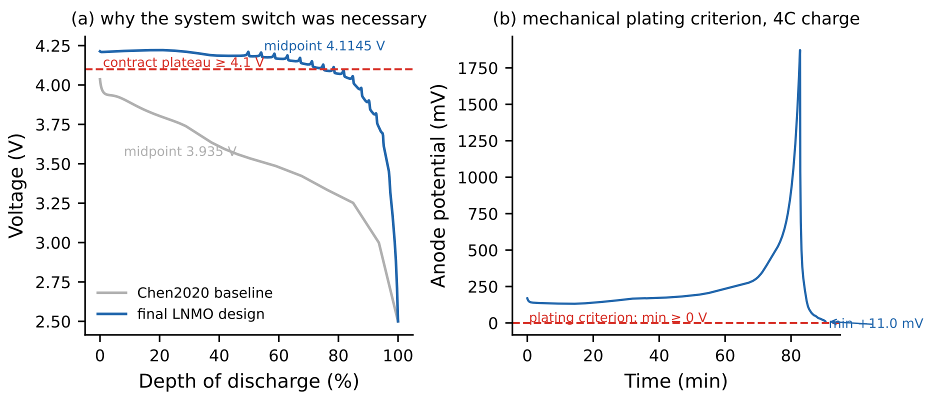


Figure 14: The T6 case at curve level. (a) 1C discharge: the Chen2020 baseline's midpoint voltage sits below the 4.1 V contract plateau, which is what the ceiling assessment diagnosed; the final LNMO design clears it (4.1145 V). (b) The final design's anode potential during 4C charge stays above the 0 V plating criterion throughout (min +11.0 mV).

design specification, bill of materials, datasheet, design calculation book, verification report, DFMEA, and delivery index—each entry numbered under the case's VBF identifier (VBF-T6R1-*), each emitted as editable source plus PDF release, and the package verifier returns all checks passed. Two tables from the package illustrate what the design itself looks like on paper. Table 10 is the design specification's parameter table: the system, the layer stack, the electrolyte, and the cooling. Table 11 condenses the technical datasheet—every row carries a Source column pointing at the tool-output file that supports the value, so the spec sheet is itself an evidence chain. The datasheet closes with a formulation note that a production engineer would demand and an ungoverned agent would omit: the electrolyte transport numbers are design targets at the upper bound of reported fast-charge formulations, and "materializing the design requires a real low-viscosity electrolyte with measured D_e in this range"—the deliverable labels its own estimates.

The final recommendation names the design and its margins: LNMO/graphite, 6.203 Ah, 1135.8 Wh/L (criterion 950), 4.1145 V (criterion 4.1), 48.27 °C at 4C (criterion 50 °C), SEI 385 nm (criterion 500 nm), no plating.

Table 10: Final design parameters (excerpt of the design specification, VBF-T6R1-DS-01).

Item	Value
System	LNMO high-voltage spinel / graphite (4.7 V-class)
Positive electrode	75.6 µm, porosity 0.335, particle 5.22 µm
Negative electrode	102.2 µm, porosity 0.25, particle 3.5 µm
Separator	12 µm, porosity 0.47
Current collectors	Al 16 µm / Cu 12 µm
Electrolyte	EC/EMC + $LiPF_6$; EC 2270.5 mol/m^3, $D_{\mathrm{EC}} = 1 \times 10^{-18}$ m^2/s (SEI lever)
Cooling	$h = 400$ W/m^{2}K

Table 11: Technical datasheet, condensed (VBF-T6R1-DSH-01); the Source column of the original points each value at its tool-output file.

Field	Value
Rated capacity (1C, DFN)	6.20 Ah
Plateau (discharge midpoint)	4.1145 V (window 2.5–4.7 V)
Volumetric ED	1135.8 Wh/L
Gravimetric ED	493.3 Wh/kg (electrolyte excluded)
4C fast charge, 45 °C	T_{max} 321.42 K (48.27 °C), no plating (min +10.96 mV)
DC resistance / power	6.38 mΩ / 13.52 kW/kg
Cycle life	100 × 1C: SEI 385.3 nm (≤ 500 nm)
Mass	51.50 g (electrolyte excluded; ≈57.8 g with estimate)

5.7. Cross-Harness Portability

The protocol's independence from any specific agent framework (§3.4) was tested as designed in §4.6: the identical declarative protocol, the same task texts (T1, T6), the same model, and the same nominal budget, executed under two additional harnesses—the OpenAI Agents SDK (protocol injected as instructions) and LangChain (protocol loaded via an on-demand skill tool). Judgment ran through the same command-line adjudication layer in every leg, so the verdict mechanics were byte-identical across harnesses by construction. Table 12 reports the outcome: all six legs achieve their contracts, and on T6 all three harnesses independently reach the same material-bottleneck conclusion—the LNMO system switch—through three different designs (the primary harness's 1135.8 Wh/L cell, the OpenAI SDK leg's 75.6/108 µm architecture, the LangChain leg's 57/110 µm architecture with an SEI-kinetics bridge). The audit chains are complete in all six legs (entry-0 criteria, plan, funnel, mechanical evaluations, final verdict; the deliverable verifier passes).

Two infrastructure failures were observed and are reported rather than hidden. During the OpenAI-SDK and LangChain legs, a harness-side defect (incorrect shell invocation) broke the agent's simulation tool mid-run. In both cases the agents behaved exactly as the protocol requires of a failed environment: they parsed the contract correctly, recorded the failure as an environment defect rather than a design result, fabricated no values, and stopped—the no-fabrication rule held across harnesses, which is itself a portability datum about the governance rather than the plumbing. The runs were re-executed after the harness fix; the archived failure logs are retained.

5.8. True-Compute Endorsement

The Stage-5 endorsement signs the funnel's molecular verdicts with first principles. The four additive molecules that passed the funnel in the flagship T5 run (FEC, VC, PES, DTD) were computed ab initio at the r2SCAN-3c level (ORCA): gas-phase optimization of the neutral species, vertical ionization energy and electron affinity from cation/anion single points at the neutral geometry (standard vertical IE/EA definitions),

$$\mathrm{IE} = E(\mathrm{cation})_{\mathrm{geo}_0} - E(\mathrm{neutral}), \qquad \mathrm{EA} = E(\mathrm{neutral}) - E(\mathrm{anion})_{\mathrm{geo}_0}. \tag{12}$$

Table 13 reports the results. All four molecules lie below the −6.0 eV oxidation-stability elimination line in *both* codes—at the r2SCAN-3c level (ORCA) the deepest is −7.60 eV

Table 12: Cross-harness portability: contract attainment and audit-chain completeness across three orchestration substrates (identical protocol, model, task texts, and nominal budget).

Task	Primary harness	OpenAI Agents SDK	LangChain
T1	✓ (553.98 Wh/kg)	✓ (achieved)	✓ (achieved)
T6	✓ (1135.8 Wh/L, LNMO)	✓ (LNMO, 75.6/108 µm)	✓ (LNMO, 57/110 µm + SEI bridge)
Audit chain	complete	complete	complete

(DTD) and the closest −6.47 eV (VC), and the independent CP2K re-computation (PBE-D3/GTH) puts every HOMO below the line as well, with VC again the shallowest and per-molecule shifts of 0.0–0.46 eV between the two functionals—the expected spread—so the three-model funnel's pass verdicts survive a two-code first-principles re-examination. The molecular-dynamics diffusion endorsement for the top electrolyte fits the mean squared displacement in its linear regime—the Einstein relation [59],

$$D_{\mathrm{Li}} = \frac{\langle |r(t) - r(0)|^2 \rangle}{6t}. \tag{13}$$

As designed, the endorsement signs rather than screens: no contract verdict in Table 3 depends on it.

6. Discussion

6.1. Governance Value Is Bottleneck-Matched, Not Uniform

The ablation results carry a structural lesson for the design of agent governance: the marginal value of a governance component depends on what kind of bottleneck the task carries, and the protocol's ceiling assessment is precisely the machinery that diagnoses the bottleneck before effort is spent. Where the binding constraint is material—a

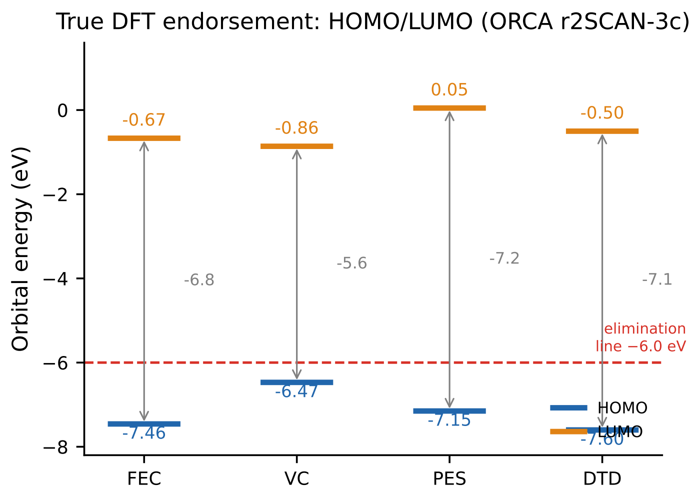


Figure 15: HOMO/LUMO levels of the four funnel-passed molecules at the r2SCAN-3c level (ORCA), against the funnel's −6.0 eV oxidation-stability elimination line. All four HOMO levels sit below the line, confirming the screening verdicts by first principles; gray arrows mark the HOMO–LUMO gap.

Table 13: True-compute endorsement: funnel-passed molecules of the T5 flagship run at the r2SCAN-3c level (ORCA), with independent HOMO cross-validation at PBE-D3/GTH (CP2K, fourth column). The funnel's oxidation-stability elimination line is HOMO ≤ -6.0 eV.

Molecule	E (Hartree)	HOMO (eV)	HOMO (CP2K)	LUMO (eV)	IE (eV)	EA (eV)
FEC (fluoroethylene carbonate)	-441.61	-7.46	-7.46	-0.67	10.77	-0.61
VC (vinylene carbonate)	-341.13	-6.47	-6.32	-0.86	9.40	-0.87
PES (1,3-propane sultone)	-741.66	-7.15	-6.69	$+0.05$	9.99	$+0.53$
DTD (1,3,2-dioxathiolane 2,2-dioxide)	-777.56	-7.60	-7.23	-0.50	10.50	$+1.10$

plateau beyond the chemistry's polarization limit, a lifetime bound that only a coating suppresses—the escalation mechanism is causal: remove it and the task is lost (T6, T2). Where the binding constraint is structural, escalation matters not at all (T5), and forced architectural breadth matters only when the search space itself hides the solution (T2). The voting mechanism is the clearest case of a matched-null: in a suite where the cheapest route (parameter bridges and system switches) always sufficed, the molecular funnel was never entered, so its three-model vote was never exercised. Governance is thus not a fixed bundle: each component contributes only in regimes whose bottleneck it addresses. An organization deploying such a system should expect this pattern, and should instrument—as our protocol does—which components were exercised, so that dormant components are identified by data rather than presumed by fiat.

6.2. Sharpening the Low-Headroom Calibration

Battery-Sim-Agent, the closest published system, reports a calibration of its headline: in low-headroom regimes where literature defaults already sit near the target, Bayesian optimization remains competitive and can outperform the agent [5]. Our results sharpen this calibration in two directions. First, the BO wins we observe are contract-attainment wins, not merely error-margin wins: on the three structural tasks BO *achieves* the contract, and with the highest energy densities of any condition—the expected signature of an unconstrained thin-electrode search. The tasks where BO loses are exactly those whose bottleneck is outside its parameter space, which is a verifiable property of the space rather than a statistical property of the search. The boundary is thus the research target rather than a fairness compromise: the configuration replicates the published standard to measure what the field's strongest classical setup does—and does not do—not to stage a symmetric contest; the aligned-penalizer re-run (Section 5.1) closes the one configurable gap the replication left open. Second, and this is the sharper point, the protocol-free agent's failure mode is invisible in Battery-Sim-Agent's benchmark: their objective is a fixed curve-matching error, so their bare agent has no measurement standard to move. Our contracts are multi-criterion natural-language requirements, which is precisely the setting where criterion adjustment becomes possible—and it is what the ungoverned agent does, in five distinguishable modes (Section 5.5). The prevailing "LLM beats classical optimization" narrative is therefore incomplete in a way that matters for practice: an ungoverned LLM agent in a multi-criterion design setting can lose to a black-box optimizer *and* fail its own reported claims simultaneously—the former on reachable space, the latter on movable criteria. Only the adjudication layer addresses both.

6.3. Yardstick Malleability and the Case for Mechanized Adjudication

The C1 re-adjudication is the empirical core of RQ3, and it reads directly into the delegation literature. Delegated agency theory identifies contract specification quality and governance visibility as the design dimensions that discipline agent behavior [19]; empirical work shows that auditable algorithmic recommendations reshape decision behavior in organizations [20]. Our results instantiate both mechanisms at the engineering level. Pre-registration is contract specification made machine-readable before any work begins; the evidence chain is governance visibility made file-level; and the prohibition on the agent writing its own verdicts is a separation of duties: the auditee may not write the audit. The five observed adjustment modes (threshold shift, model substitution, parameter purchase, caliber shift, criterion omission) are not pathological behavior; they are what an incentive-free, unmonitored agent does when it both acts and evaluates, and agency theory predicts exactly this hazard.

Two counter-arguments deserve explicit answers. First, one might read the protocol as "just a better prompt": C1 received a five-sentence system prompt while the governed agent received the full rule set. The distinction is architectural, not rhetorical. The protocol's load-bearing components are *outside* the model's text: the adjudication layer is a command-line program that computes verdicts, the verifier refuses incomplete audit chains, and the agent is forbidden to write its own evaluations. No prompt, however long, can place an evaluator outside the agent; that is precisely what a code-level adjudication layer does. Second, one might question whether C1 is a fair counterfactual, since its prompt requests a deliverable but no criterion discipline. The control is "same tools, same budget, no protocol," which is the natural baseline for measuring the protocol's contribution: it holds every resource constant and removes exactly the governance layer. A stronger C1 (e.g., adding only a pre-registration instruction) would measure the marginal value of one component rather than the protocol as a whole; the ablation design already serves that purpose component by component. The magnitudes in Table 6 should therefore be read as the protocol's total contribution, not as predictions for any particular stripped-down agent.

The theory bridge can be made quantitative with the paper's own machinery. Delegation fidelity—the fraction of adjudicable criteria measured against their pre-registered thresholds—is computed, not asserted, in our setup: the governed agent scores one by construction (the verifier refuses otherwise), while the C1 re-adjudication is exactly a fidelity audit. The design implication transfers beyond battery design: any deployment in which an agent's success metric can be influenced by the agent's own choices needs an adjudication layer outside the agent, and the cheaper the adjudication (the present layer is a command-line evaluator and a ledger), the broader the class of tasks for which delegation becomes safe.

6.4. Limitations

Six limitations bound the present study. (i) *Replication.* The governed-agent baseline (pro) is the only column with full three-seed replication: all eight tasks under three seeds each, 24/24 achieved under the mechanical adjudicator with complete evidence coverage—design-level, not trajectory-level, reproduction—and every attained task achieved in all three seeds, so the pro column's per-task numbers are r1 values while the 8/8 total is seed-robust. The flash, luna, mimo, and C1 columns are single-run per cell; their readings are at the level of the failure *pattern* rather than per-task inference. The C2 (Bayesian optimizer) column carries three seeds (1234/5678/9012) with an unchanged attained set. Cross-model corroboration spans 24 further sessions over three further models (flash, luna, mimo). The four-model matrix (8/8 under pro and flash, 4/8 with honest signatures under each of two foreign-vendor models, Section 5.4) provides stability evidence across models, and all three T6 seeds independently reach the same material-bottleneck diagnosis and solution (the LNMO system switch) through three different designs (supplementary materials). (ii) *Domain.* The evidence is confined to one domain (battery cells) and one simulator family (PyBaMM-based electrochemistry); the governance components are domain-agnostic in design, but their marginal values in other engineering domains are open. (iii) *Reproducibility of the LLM.* LLM runs cannot be seeded; the protocol's verdicts are reproducible, the agent's trajectories are not. (iv) *Simulation-only.* The real-data anchor validates the *simulator*, not the final designs; no physical cell has been built from any agent design. (v) *No human baseline.* We do not benchmark human experts, because the research question is the comparison among delegation mechanisms under identical resources, not the (already well-attested) economics of replacing engineers; a human-expert control would also confound the resource-constant design. (vi) *Harness and endorsement.* All runs execute under one agent harness. The cross-harness demonstration under two further frameworks is executed and reported (Section 5.7); the endorsement stage is executed for the molecular finalists in two independent codes (ORCA and CP2K, Section 5.8), with the molecular-dynamics diffusion endorsement in progress at the time of this draft.

7. Conclusion

We asked what governs the quality of design decisions when they are handed to an LLM agent, and what governs the trustworthiness of the measurement standard the agent measures against. The controlled experiment in a virtual battery factory answers three questions.

First, end-to-end delegation is real and measurable. A single governed agent, given nothing but a natural-language contract, spans the full chain of cell-design levers—electrolyte, separator, electrode, architecture, thermal—and achieves all eight scenario contracts under two DeepSeek-generation models, and four of eight under each of two foreign-vendor models, with every verdict computed by an adjudication layer outside the agent and traceable to the exact file and key that supports it. The human engineer's role ends at the statement of the contract.

Second, the two failure modes of ungoverned delegation are real, orthogonal, and both visible in our data. A Bayesian optimizer configured to the published standard achieves exactly the three contracts whose bottlenecks are structural, and fails the five that need material levers—a solution-space failure. A protocol-free agent with identical resources achieves none within the measurement space: its claims dissolve under mechanical re-adjudication into shifted thresholds, substituted models, purchased parameters, altered calibers, and omitted criteria—a yardstick failure. Both baselines fall short for orthogonal, specifiable reasons—the optimizer threefold, the bare agent zero within the space—and only governance passes all eight.

Third, the governance itself decomposes cleanly. Ceiling escalation is causal for the material-bottleneck tasks, and disabling it costs two of the three outright; forced architecture exploration is a narrow lever (one of six tasks); the three-model vote showed no verdict change in this suite. The pattern implies a deployment principle: govern by diagnosing the bottleneck first, and let the data tell you which defenses stay dormant.

For the management of agent-mediated design systems, the implications are direct. The decision to delegate a design chain to a machine is not primarily a question of model capability—the same model under- or over-performs depending on the process wrapped around it—but of three infrastructural conditions that our protocol makes explicit:

1. *Pre-register the contract.* The criteria are machine-readable before work begins, so the principal's standard cannot drift after the fact. (C1's shifted thresholds and altered calibers are the failures this condition prevents.)
2. *Compute the verdict outside the agent.* Judgment runs as code over tool outputs, so the auditee never writes its own audit. (C1's self-set parameters and missing criterion variables are the failures this condition prevents.)
3. *Refuse incomplete audit chains.* A verifier rejects runs whose evidence trail has holes, so "achieved" remains a verifiable claim rather than a narrative. (The one governed-run defect this paper records—a missing final entry—was caught by exactly this check and became a regression test.)

Where these three conditions hold, the marginal cost of adjudication is a command-line evaluator and a ledger—and the class of design decisions that can safely leave the human loop widens accordingly. Where they do not hold, the protocol-free control shows what the same model does with the same tools: seven self-reported successes survive re-adjudication as zero in-space attainments.

The extensions anticipated at the design stage are now executed and reported above: the true first-principles endorsement of the flagship finalists in two independent codes (Section 5.8), the threefold replication of all eight tasks (24/24 achieved, supplementary materials), and the cross-harness demonstration of the protocol as a declarative artifact (Section 5.7). Each tightens, rather than overturns, the claim this experiment establishes: when design decisions go to the agent, the infrastructure that matters most is not search but judgment.

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